Experimental Quantum Science using cold atoms and ions in South Africa

Focus: Quantum control of Yb ions at Stellenbosch University

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2004 2006 2010 2015 2019 2025

Shaping the future of Physics Report

CSIR: Ion trapping. Hermann Uys

SU: Cooling and trapping of Rb atoms. Christine Steenkamp

SU-CSIR: Ion trapping Hermann Uys

SU & NUL: Ion trapping CM Steenkamp N Matjelo

WORKSHOP
Quantum Science and Technology across Africa

CPUT: Physics and engineering of cold Rb atoms. Kessie Govender

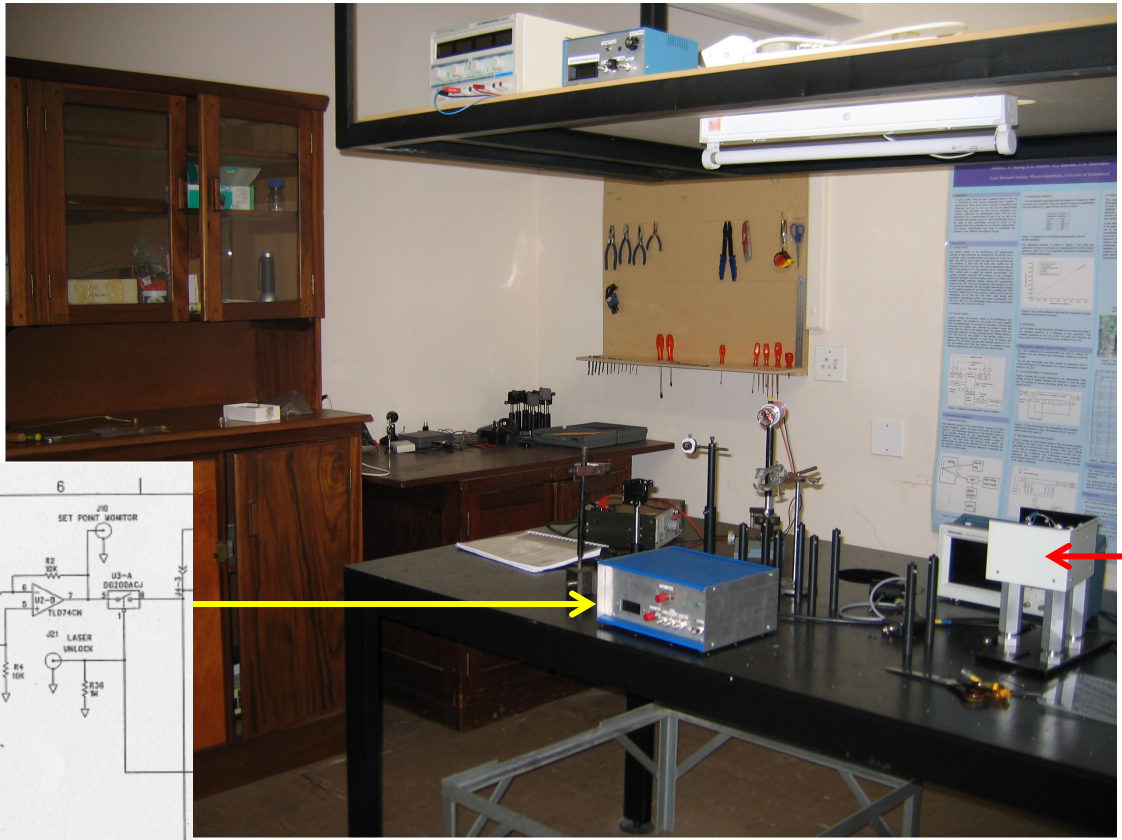
UKZN: Manipulation of ultra-cold Rb atoms using an optical nanofibers. Francesco Petruccione

[1] "Shaping the Future of Physics in South Africa" Report of the International Panel appointed by the DST, NRF, and South African Institute of Physics, April 2004.

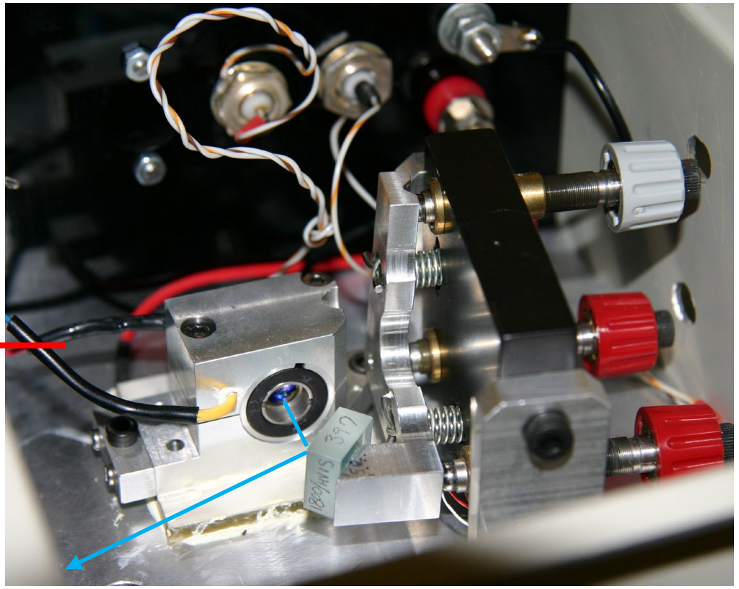
CSIR = Council for Sienctific and Industrial Research; SU = Stellenbosch University; CPUT = Cape Peninsula University of Technology; UKZN = University of KwaZulu-Natal; NUL = National University of Lesotho

Cooling and trapping Ru atoms from scratch

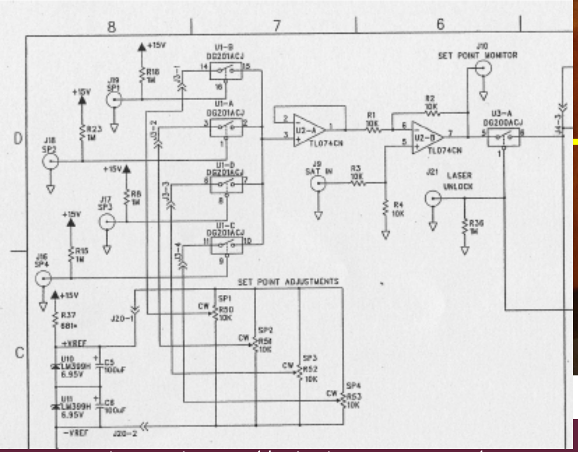
Cooling and trapping of rubidium atoms at SU



From laser diode to tunable laser



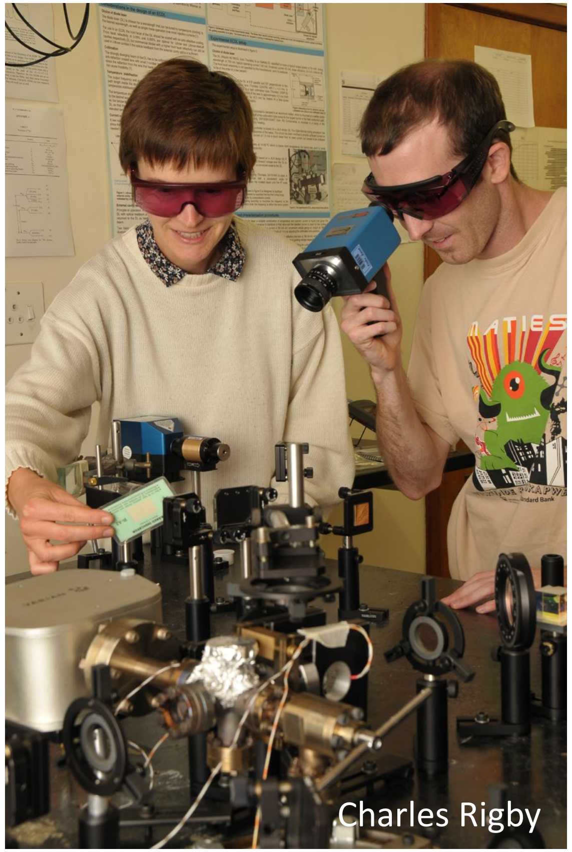
From circuit diagram to laser frequency locking

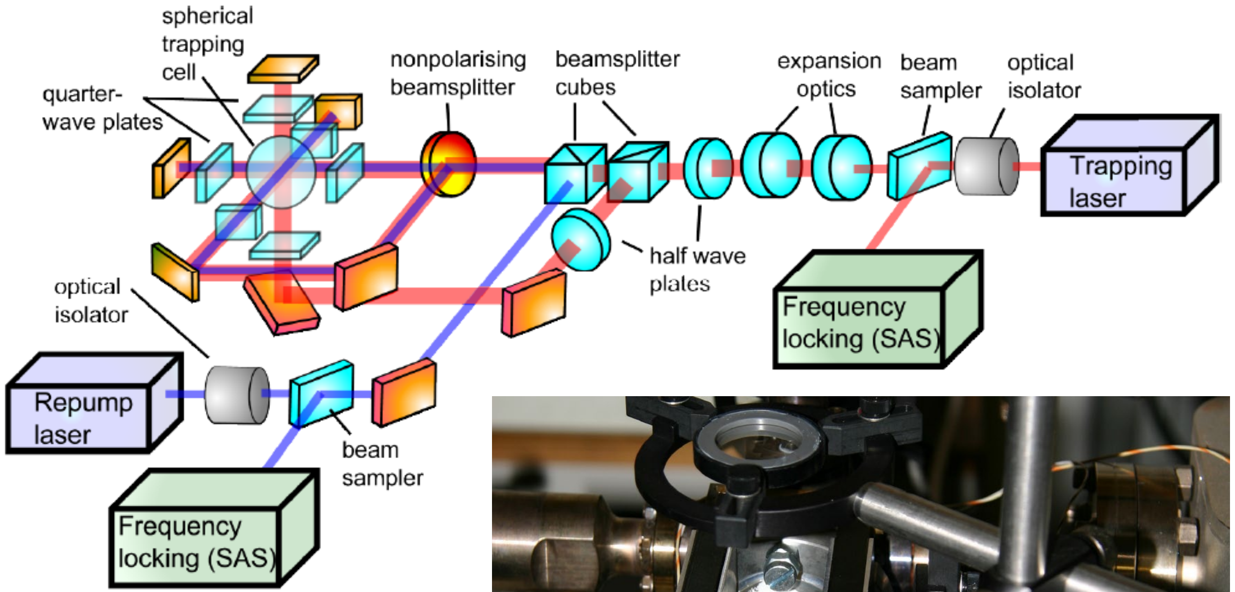


G. Nyamuda 2006, MSc thesis, https://scholar.sun.ac.za/

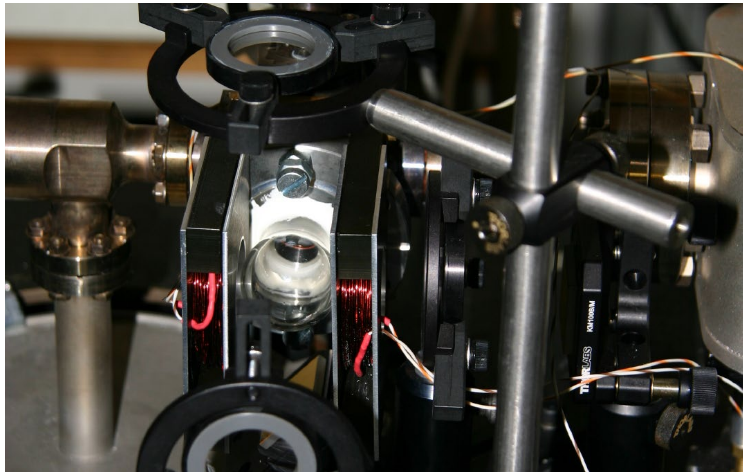
Cooling and trapping Ru atoms from scratch

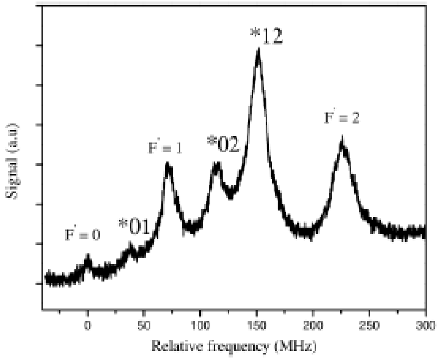
Cooling and trapping of rubidium atoms at SU





Nicolene Botha





(c) 50/50BS SAS setup

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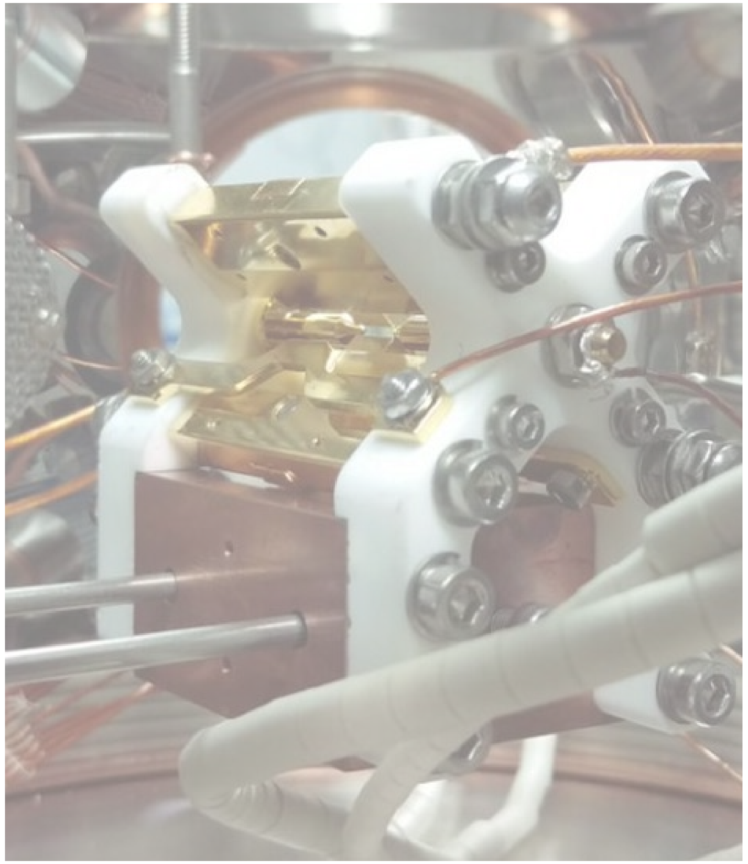
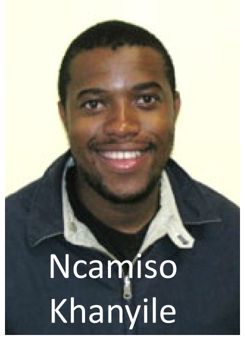
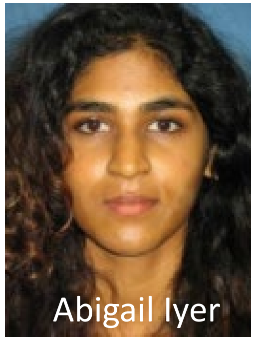
Ion trapping project at Stellenbosch

Trapping ytterbium isotopes for unsharp quantum measurements

- Ytterbium ions ^{171}Yb and ^{174}Yb
- Innovative & cost effective setup
- Aim: Realize unsharp measurement using two Yb isotopes







N Matjelo 2020, PhD dissertation; D Janse van Rensburg 2020, MSc thesis, https://scholar.sun.ac.za/

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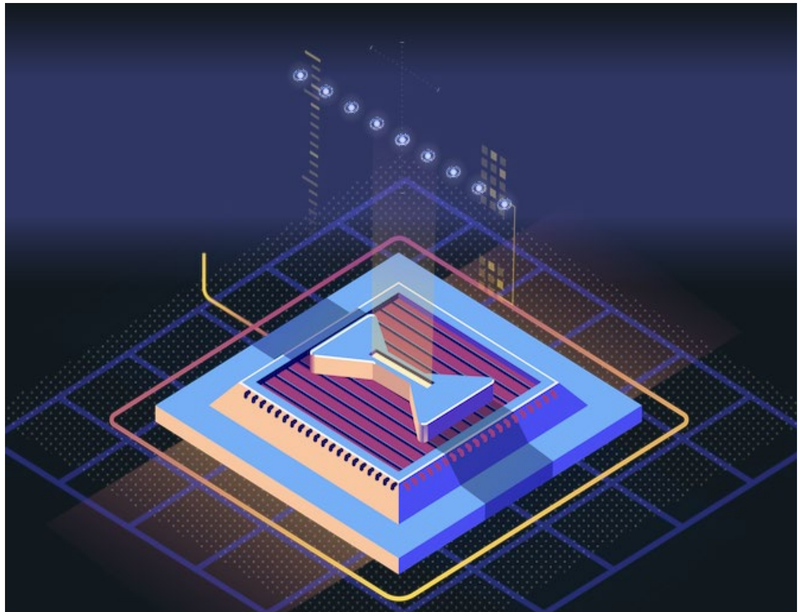
Trapped ions in quantum computing

Milestones for trapped ions in quantum computing

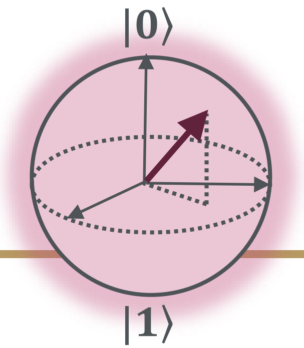
- 1980's Fundamental research with trapped ions [1]
  - Initialization, read-out of electronic states, ion crystals with multiple ions, long coherence times in vacuum
- 1990s
  - Zoller - entanglement of ions via collective motion in the trap [2]
  - Chris Monroe and David Wineland - first demonstration of two-qubit gates with $^{9}Be^{+}$ [3]
- By 2008 more than 25 research groups [1]

  Quantum protocols, more ions, miniaturization
- Commercialized: IonQ (USA), Honeywell/Quantinuum (USA), Alpine Quantum Technologies (Austria), Oxford Ionics (UK), Eleqtron (Germany),...


https://ionq.com/technology

[1] Häffner et al 2008, Physics Reports, 469, 155 and references therein; [2] Cirac and Zoller 1995, Physical Review Letters 74, 4091; [3] Monroe et al 1995, Physical Review Letters 75, 4714

Ions as qubits



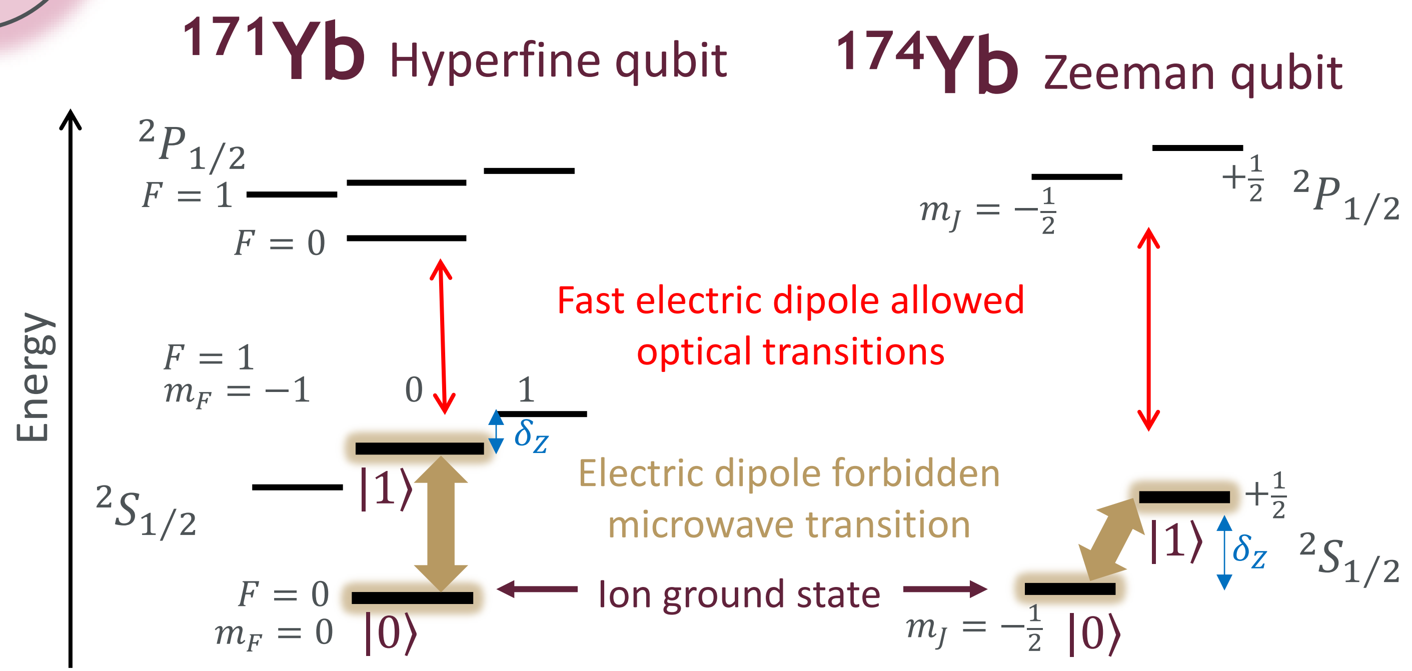
$|\psi\rangle = c_0|0\rangle + c_1|1\rangle$

An atomic ion as qubit

Qubit states $|0\rangle$ and $|1\rangle$ are chosen from the many energy eigenstates of ion.

Qubit states chosen to have long spontaneous decay lifetimes.

Fast optical transitions used to manipulate qubit state.



Zeeman splitting δ_Z in an external magnetic field.

Quantum measurements

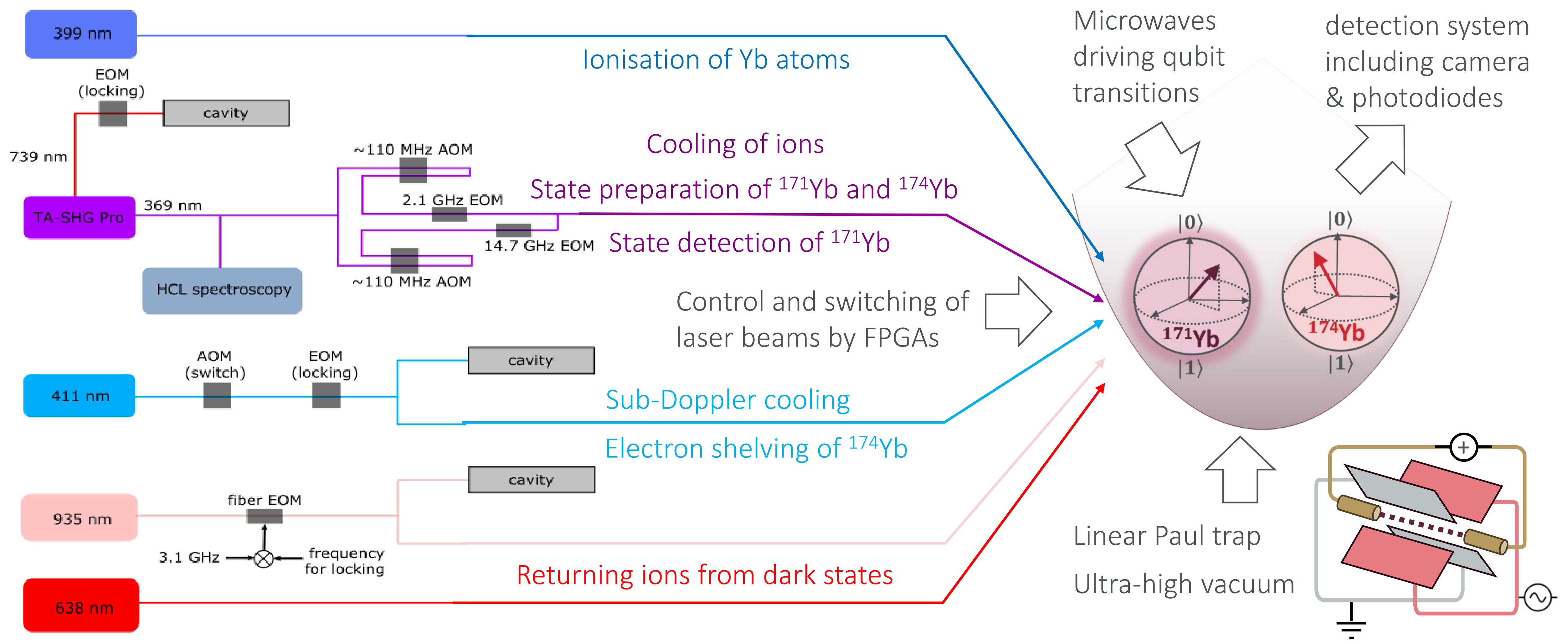
Optical transitions used to perform projective measurements (Von Neumann measurements).

Optical state detection on ion means projection to one of the energy eigenstates $|0\rangle$ or $|1\rangle$.

Overview of experimental design

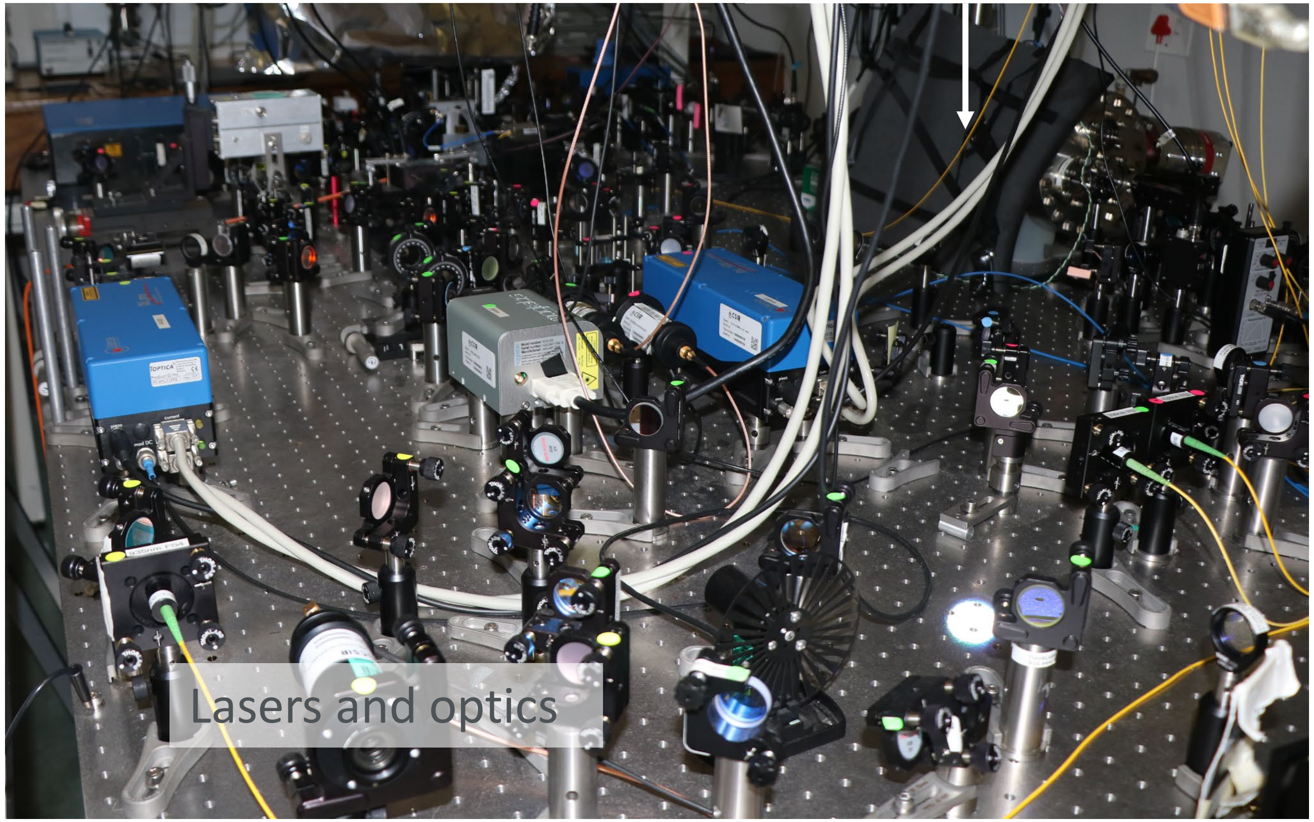
Legend:
EOM: electro-optic modulator
AOM: acousto-optic modulator
TA-SHG: Tapered amplifier second harmonic generation setup
HCL: Hollow cathode lamp containing ytterbium

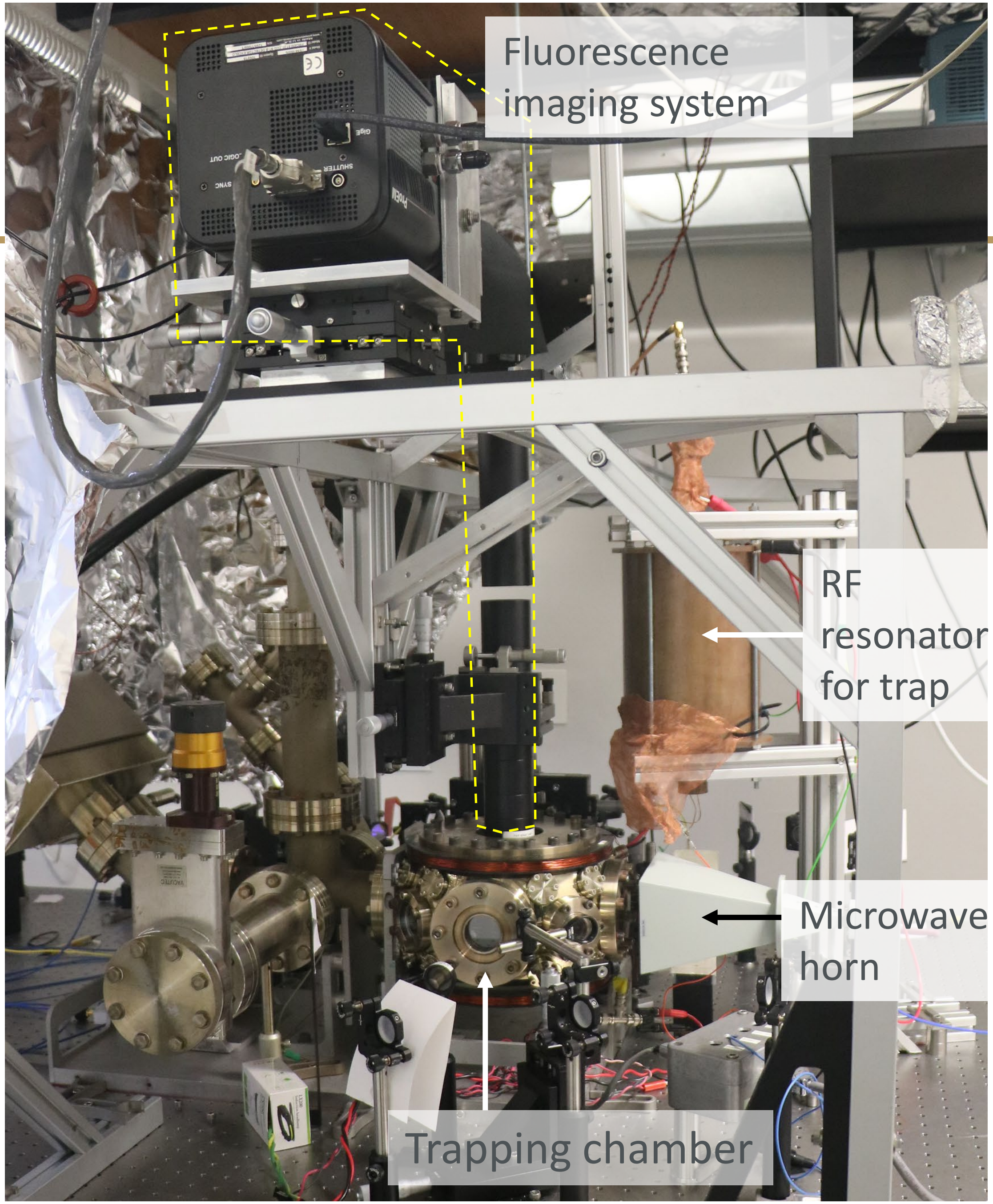
Experimental setup for trapping, cooling and controlling ^{171}Yb and ^{174}Yb qubits



Experimental setup

Chamber with cavities for frequency locking





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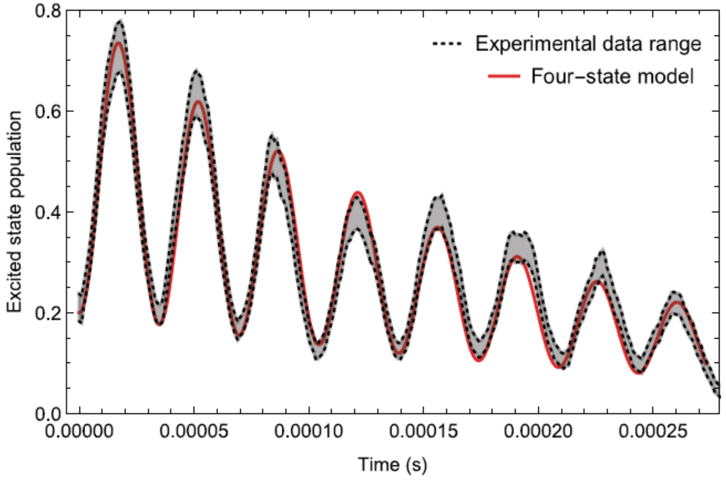
Experimental overview

First ion trapping in Africa

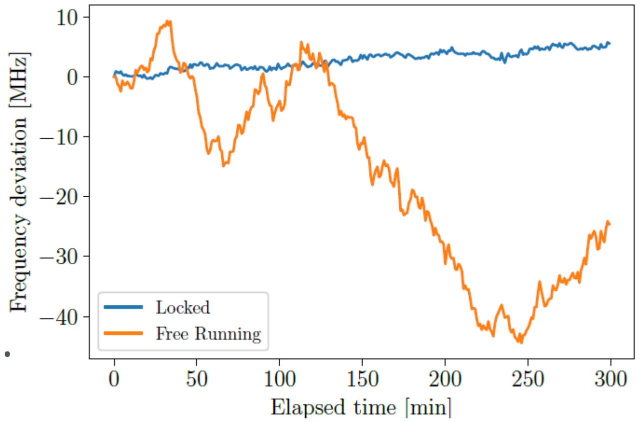


Measured Rabi oscillations of ^{171}Yb [1]

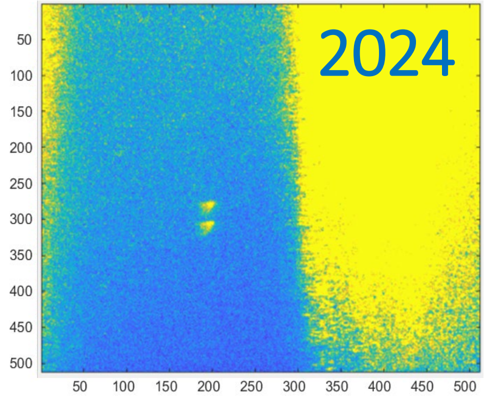
Improvements to setup

- Trap electrodes were modified.
- Vacuum was improved.
- Developed frequency locking setup for lasers [3].
- Additional laser for state detection of ^{174}Yb by electron shelving.

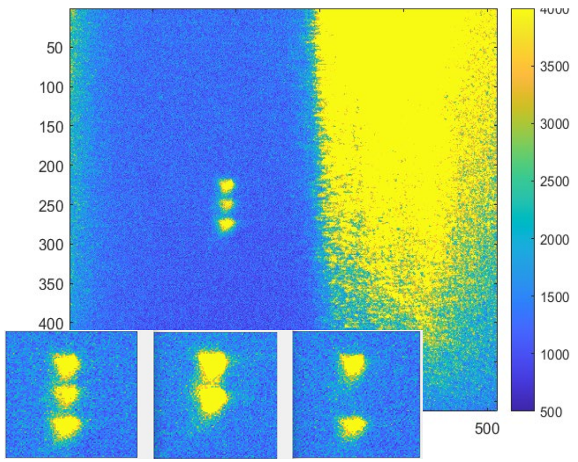
Laser frequency locked vs free-running.



Two ^{171}Yb ions in trap

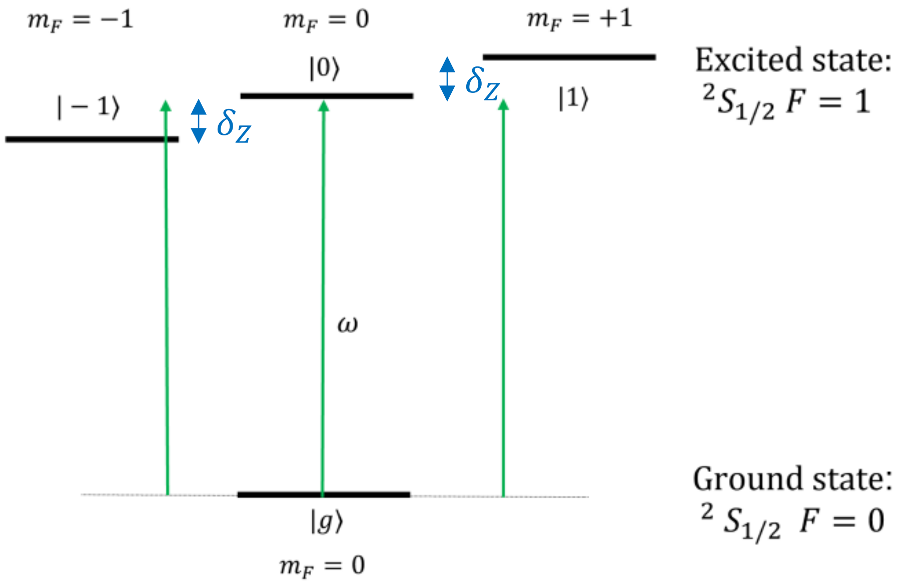


Three ^{174}Yb ions in trap



Modeling the ^{171}Yb qubit as a four-state system

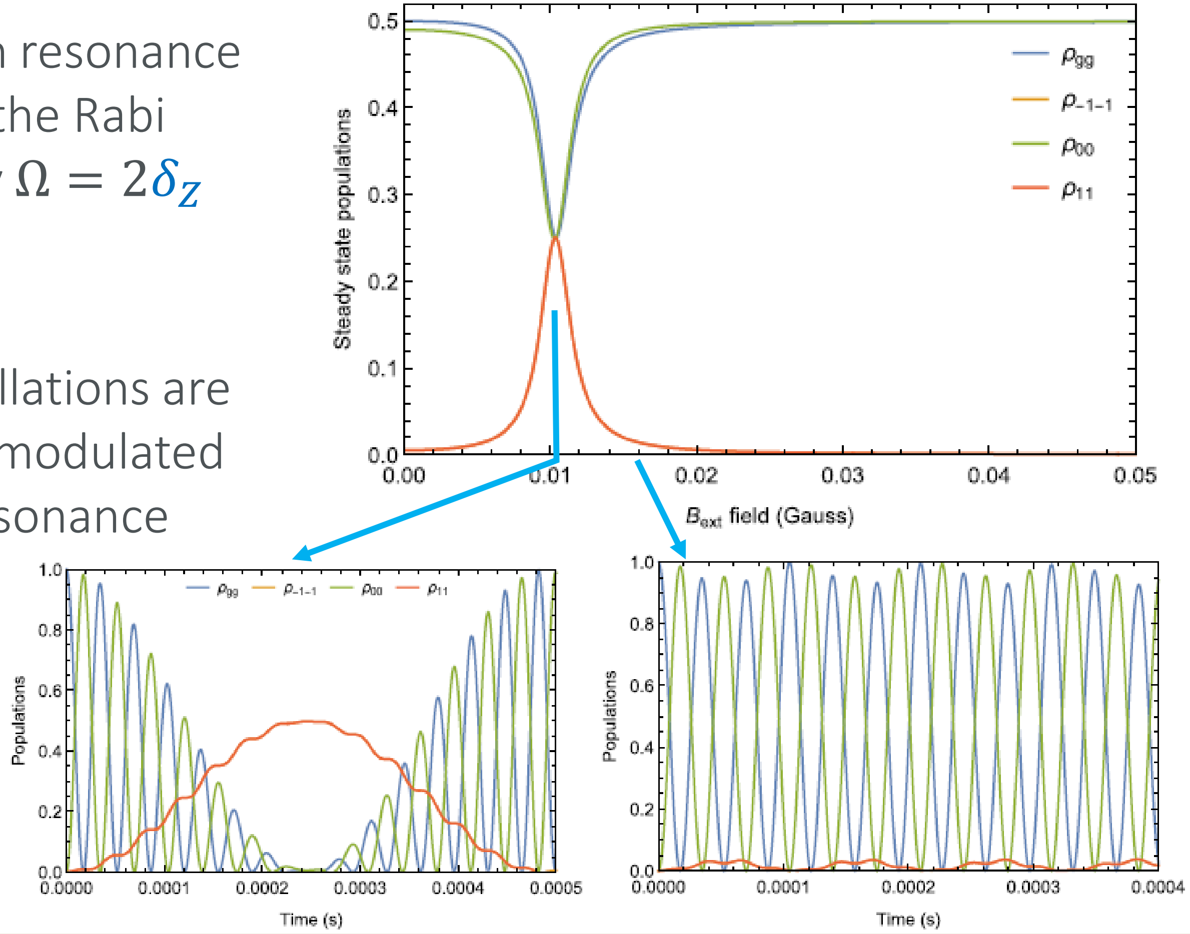
Optical Bloch equation model of the ^{171}Yb qubit in an external magnetic field [2]



Zeeman splitting δ_Z in an external magnetic field is used to lift the degeneracy of the $m_F = -1, 0, +1$ states

Saturation resonance [3] when the Rabi frequency $\Omega = 2\delta_Z$

Rabi oscillations are strongly modulated at the resonance



Simulating unsharp measurements and state estimation using trapped ytterbium ions

Oral presentation at the American Physical Society Division of Atomic, Molecular and Optical Physics Conference 2025

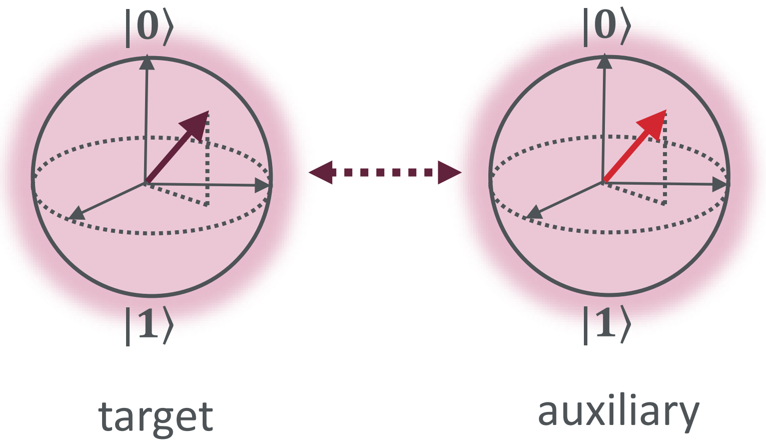
Siann Bester[1], T. Konrad[2], N.J. Matjelo[3] and C.M. Steenkamp[1]

[1]Stellenbosch Photonics Institute, Stellenbosch University, South Africa
[2]School of Chemistry and Physics, University of Kwa-Zulu Natal, South Africa
[3]Department of Physics & Electronics, National University of Lesotho, Lesotho

Introduction to unsharp measurement

Unsharp measurement

A class of **positive operator value measure** (POVM) measurements which uses a **weak unitary interaction** between the target qubit and an auxiliary qubit, so that a projective measurement on the auxiliary yields **limited information** on the target state while causing **limited change** to the target state.



Potential applications

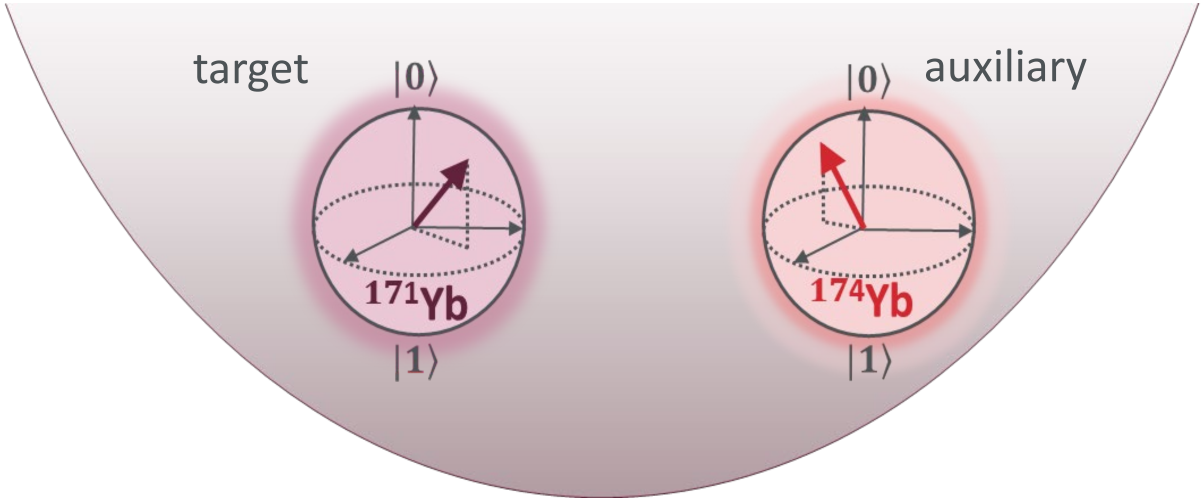
- Real time monitoring of the evolution of quantum systems [1]
- In quantum feedback protocols [2]
- Error correction [3]
- May allow for eavesdropping in a quantum key distribution networks, difficult to detect. [4]

The implementation of unsharp measurements using trapped ions has been theorised by Audretch, Konrad and collaborators [5,6].

[1] Anwer et al. 2020, Physical Review Letters, 125,080403; [2] Gillett 2010, Physical Review Letters 104, 080503; [3] Ran et al 2019, J. Phys. B 52, 165501; [4] Wath et al. 2023, Eur. Phys. J. Plus 138,54; [5] Audretsch et al. 2001, Physical Review A, 63, 052102 [6] Choudhary *et al.* Phys. Rev. A **87**, 012131 (2013)

Introduction to unsharp measurement

Implementation of unsharp measurements using trapped ions [1]

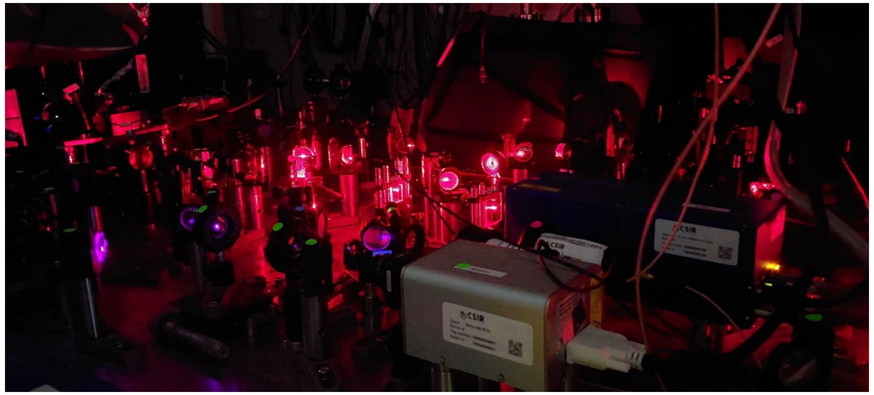


- Target and auxiliary are weakly entangled through their common motional state in the harmonic potential of the trap.
- Projective measurements are performed on auxiliary qubit.
- Weak entanglement means weak disturbance of state of target qubit.
- A series of unsharp measurements is needed to gain information, but the target qubit state is never projected or re-initialised.

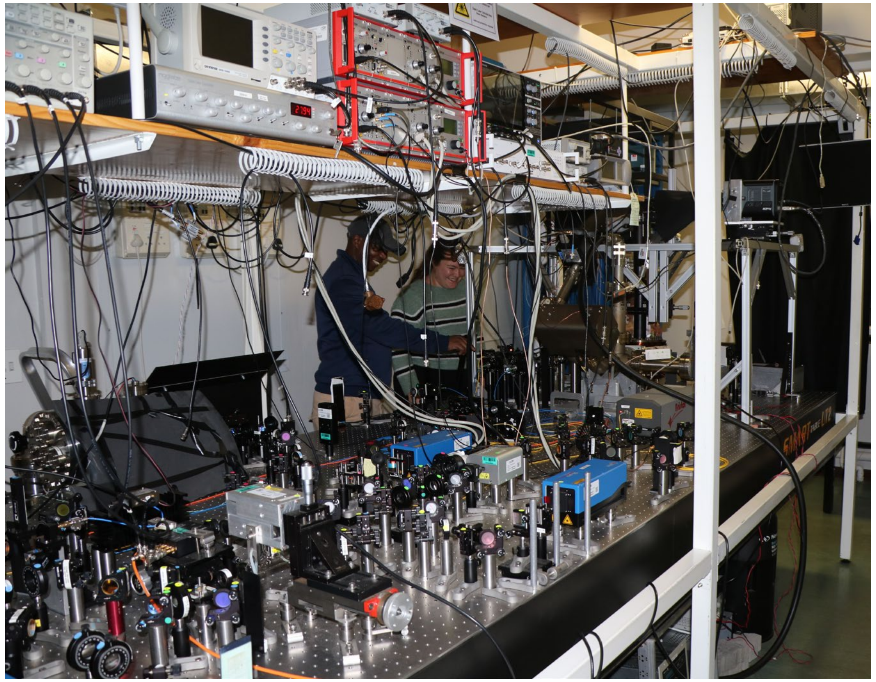
[1] Choudhary *et al*. Phys. Rev. A **87**, 012131 (2013)

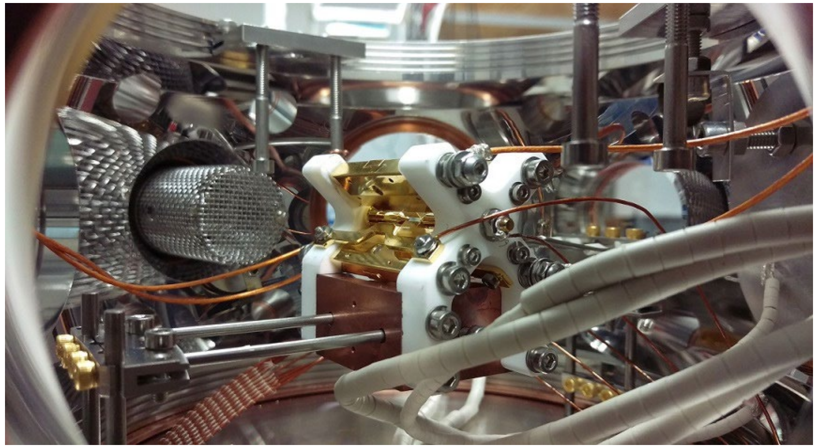
Aim of the simulation

- We investigate the feasibility of implementing unsharp measurements[3] on trapped ytterbium ions in a Paul trap from an experimental perspective.
- The study aims to track an qubit undergoing Rabi oscillations using a state estimation scheme[4].

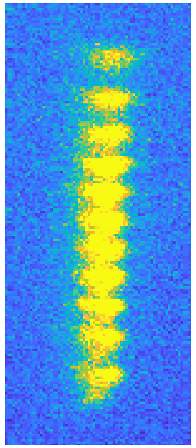


The ion trapping lab in Stellenbosch





Our linear Paul trap

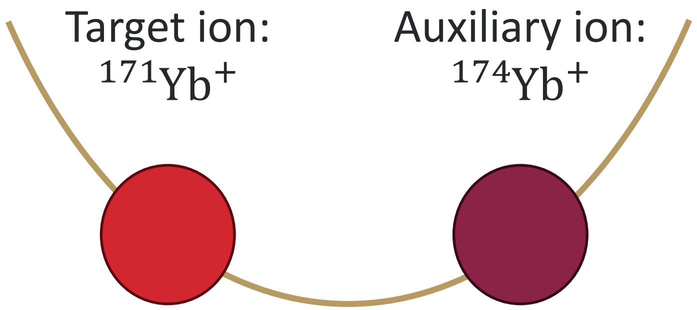


A chain of Yb-171 ions

[3] Choudhary *et al*. Phys. Rev. A **87**, 012131 (2013)
[4] Konrad and Uys. Phys. Rev. A **85**, 012102 (2012)

Unsharp measurement with two trapped ions

Two Yb ions in the same trap



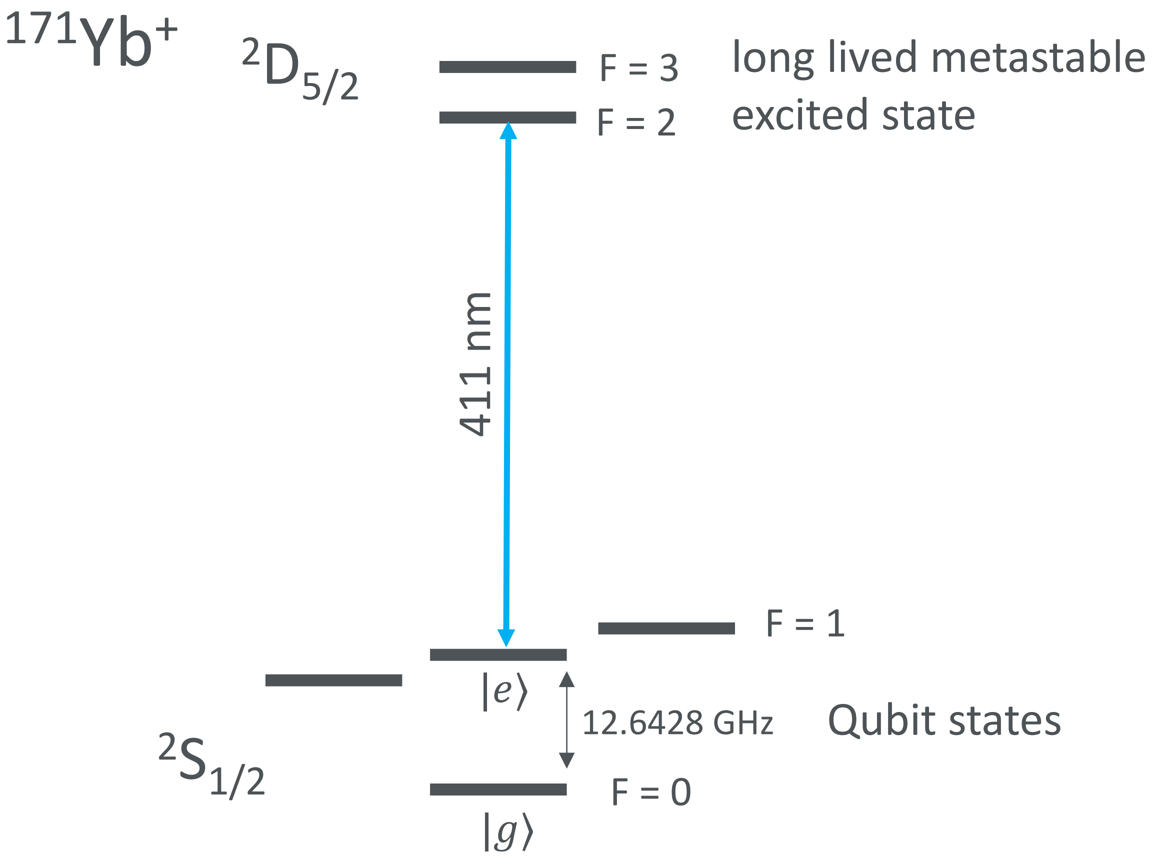
$$|\psi\rangle_{\text{initial}} = (c_1\,|g_t\rangle + c_2|e_t\rangle)\,|g_a\rangle|0\rangle$$

Target state

Motional state

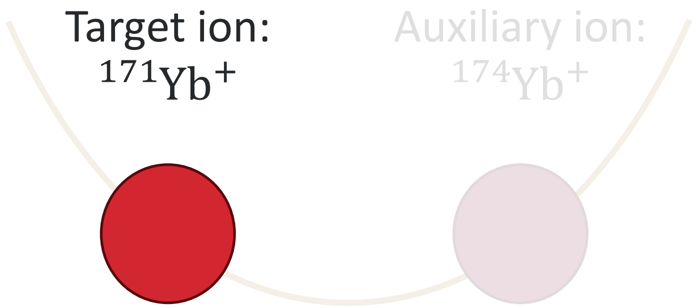
Auxiliary state

Target ion: relevant energy levels

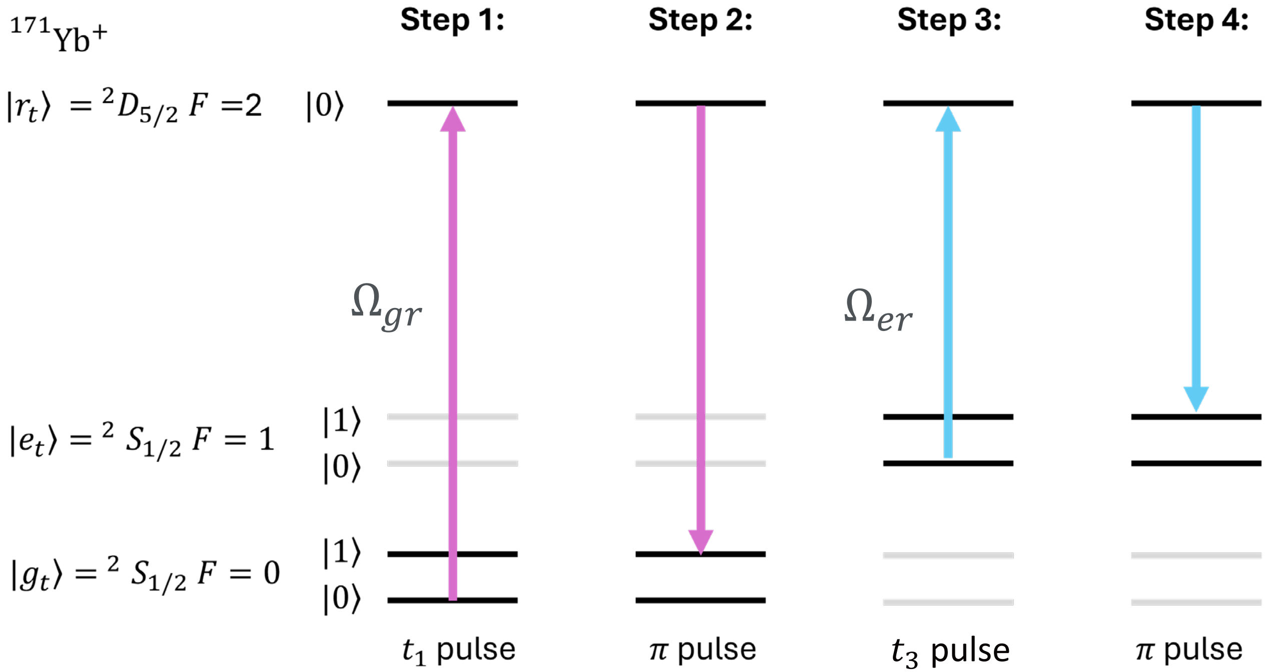


Unsharp measurement preparation scheme

Weak entanglement of target state with motional state by sequence of four pulses on target ion[3]



$$|\psi\rangle_{\text{initial}} = (c_1\,|g_t\rangle + c_2|e_t\rangle)\,|g_a\rangle|0\rangle$$

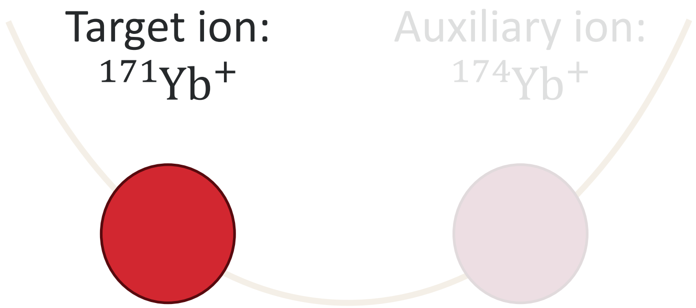


$$t_1 = \frac{2}{\Omega_{gr}}\cos^{-1}\left(\sqrt{p_0}\right) \qquad t_3 = \frac{2}{\Omega_{er}}\cos^{-1}\left(\sqrt{1-p_0}\right)$$

Projective $0 \leq p_0 \leq 0.5$ Infinitely weak

[3] Choudhary *et al*. Phys. Rev. A **87**, 012131 (2013)

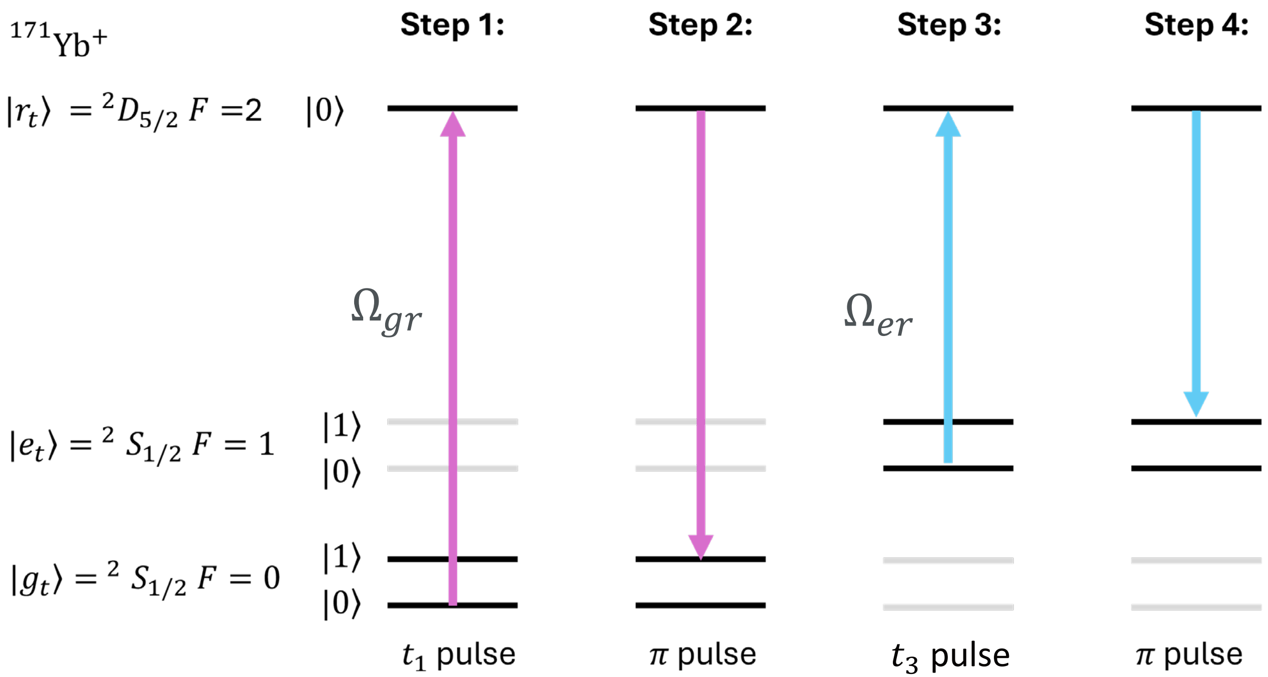
Unsharp measurement preparation scheme



$$|\psi\rangle_{\text{initial}} = (c_1\,|g_t\rangle + c_2|e_t\rangle)\,|g_a\rangle|0\rangle$$

$$|\psi\rangle_{\text{final}} = (c_1\sqrt{p_0}\,|g_t\rangle + c_2\sqrt{1-p_0}|e_t\rangle)|0\rangle|g_a\rangle + \left(c_1\sqrt{1-p_0}|g_t\rangle + c_2\sqrt{p_0}|e_t\rangle\right)|1\rangle|g_a\rangle$$

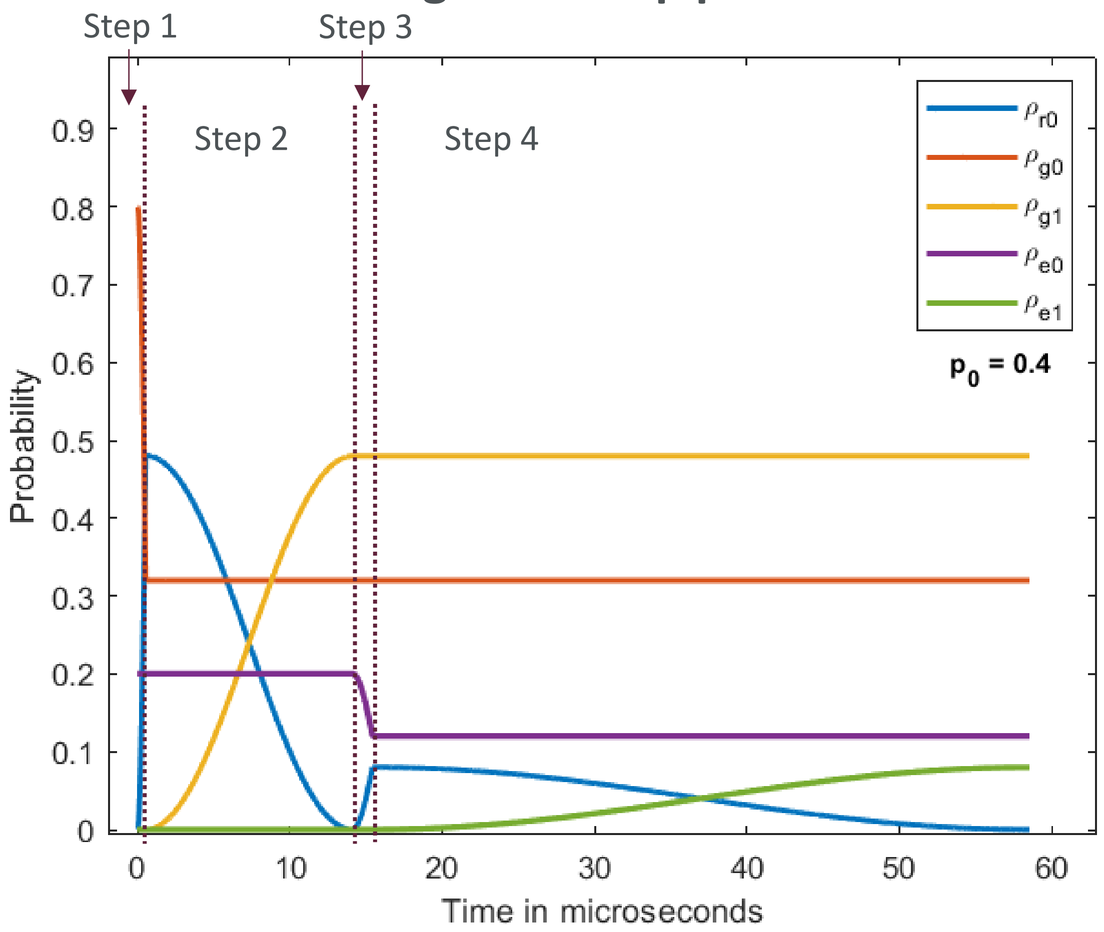
Preparation scheme[3]

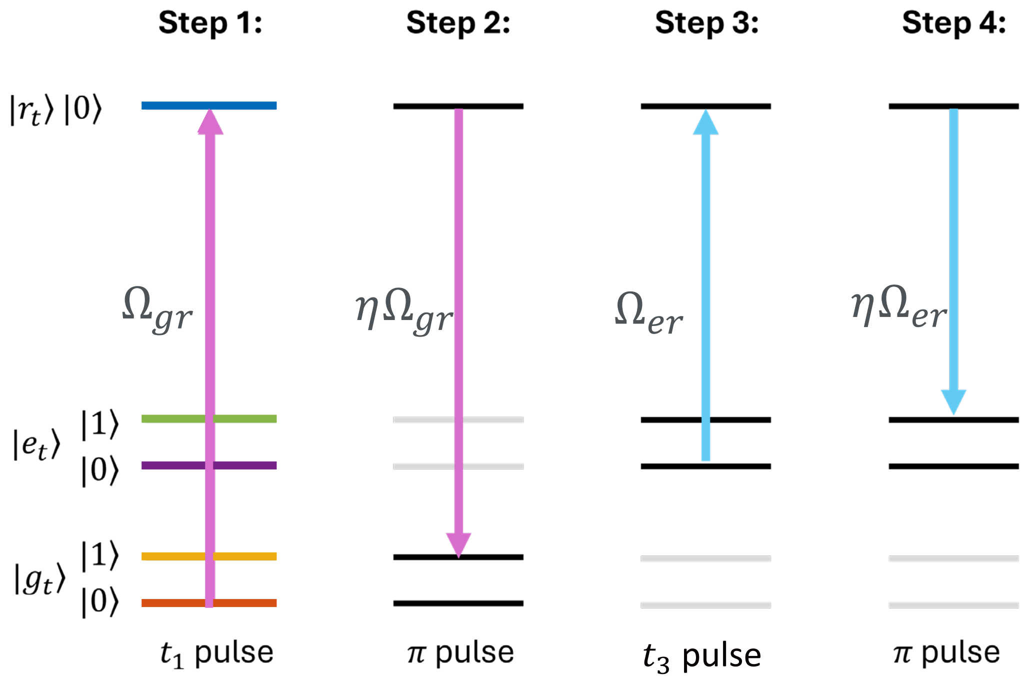


[3] Choudhary *et al*. Phys. Rev. A **87**, 012131 (2013)

Simulating an unsharp measurement

The population evolution of the 5 energy levels during the 4-step procedure



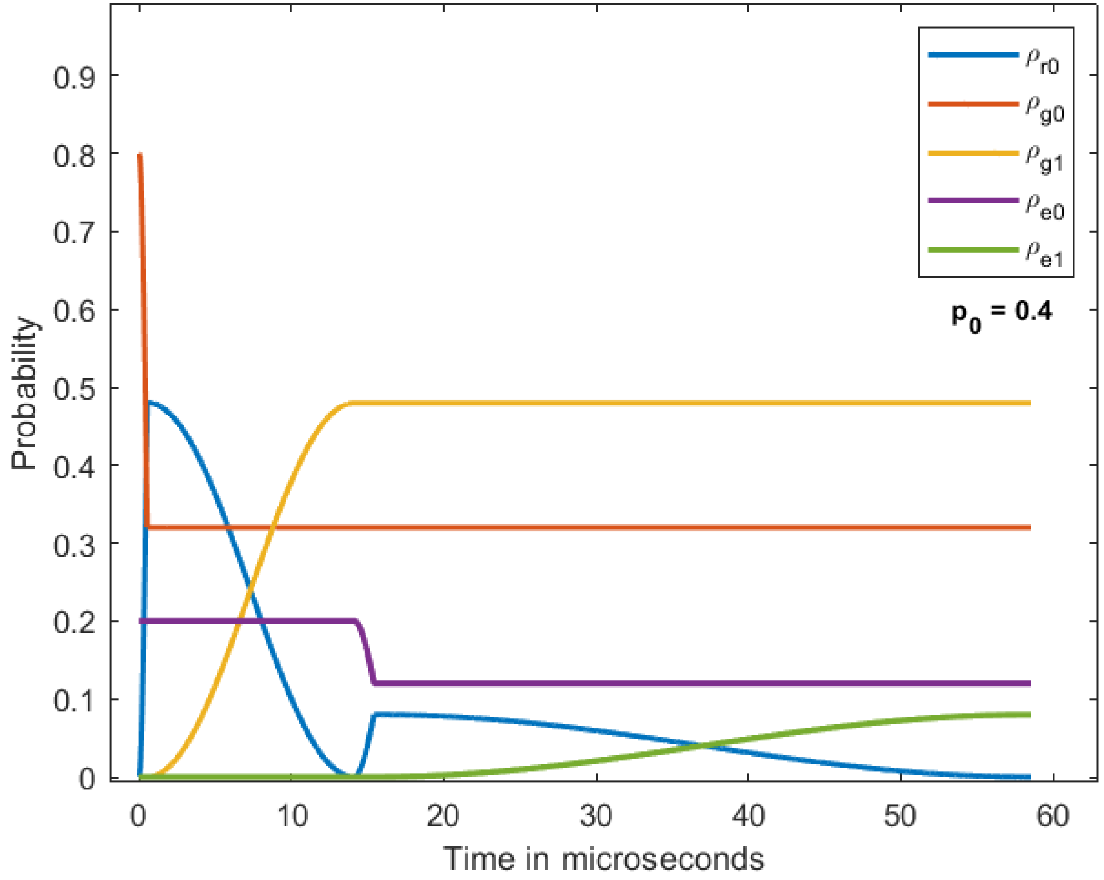


$$\text{Lamb-Dicke factor} = \frac{change\ in\ kinetic\ energy\ during\ emission}{energy\ spacing\ of\ motional\ states}$$

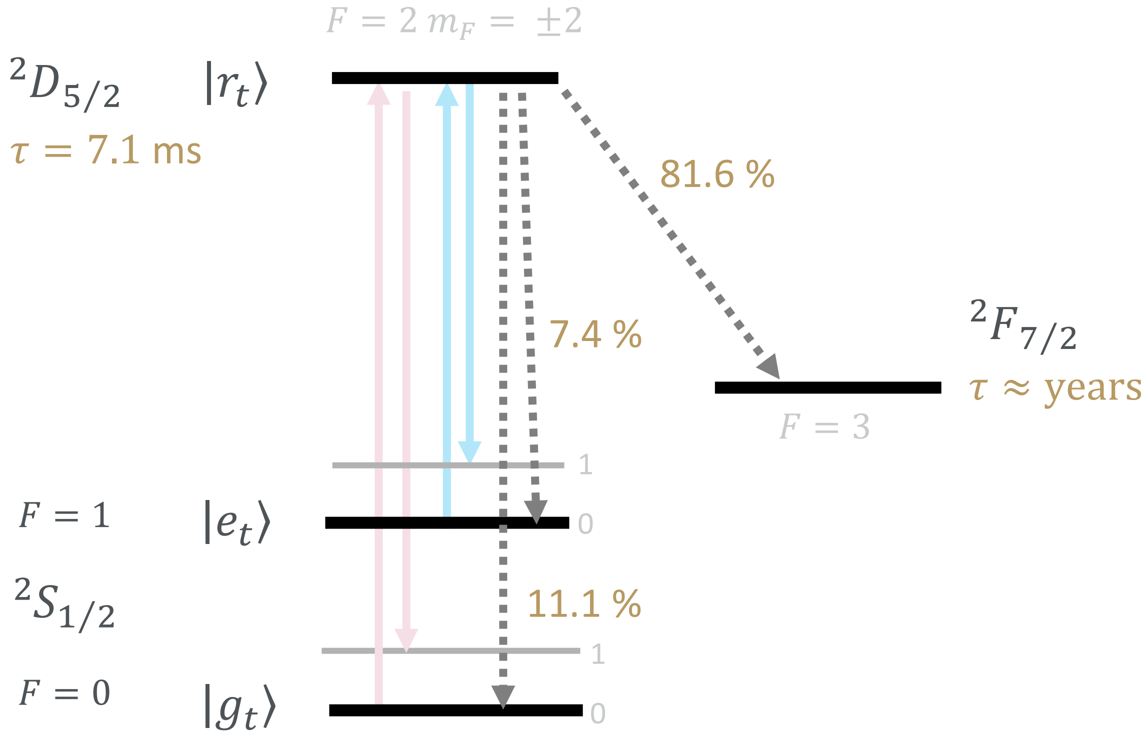
$$\eta \approx 0.07$$

Simulating an unsharp measurement with decay

The population evolution of the 5 energy levels during the 4-step procedure



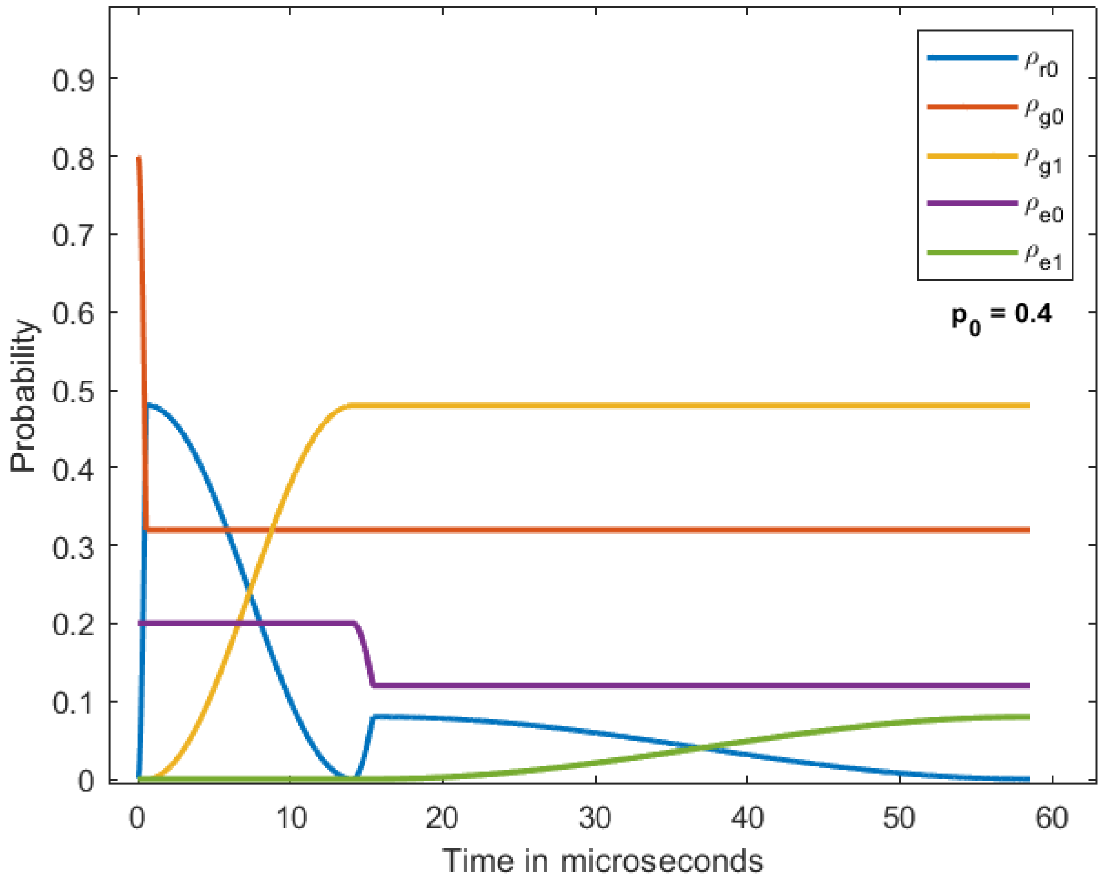
Spontaneous decay of the $^2D_{5/2}$ level[5]



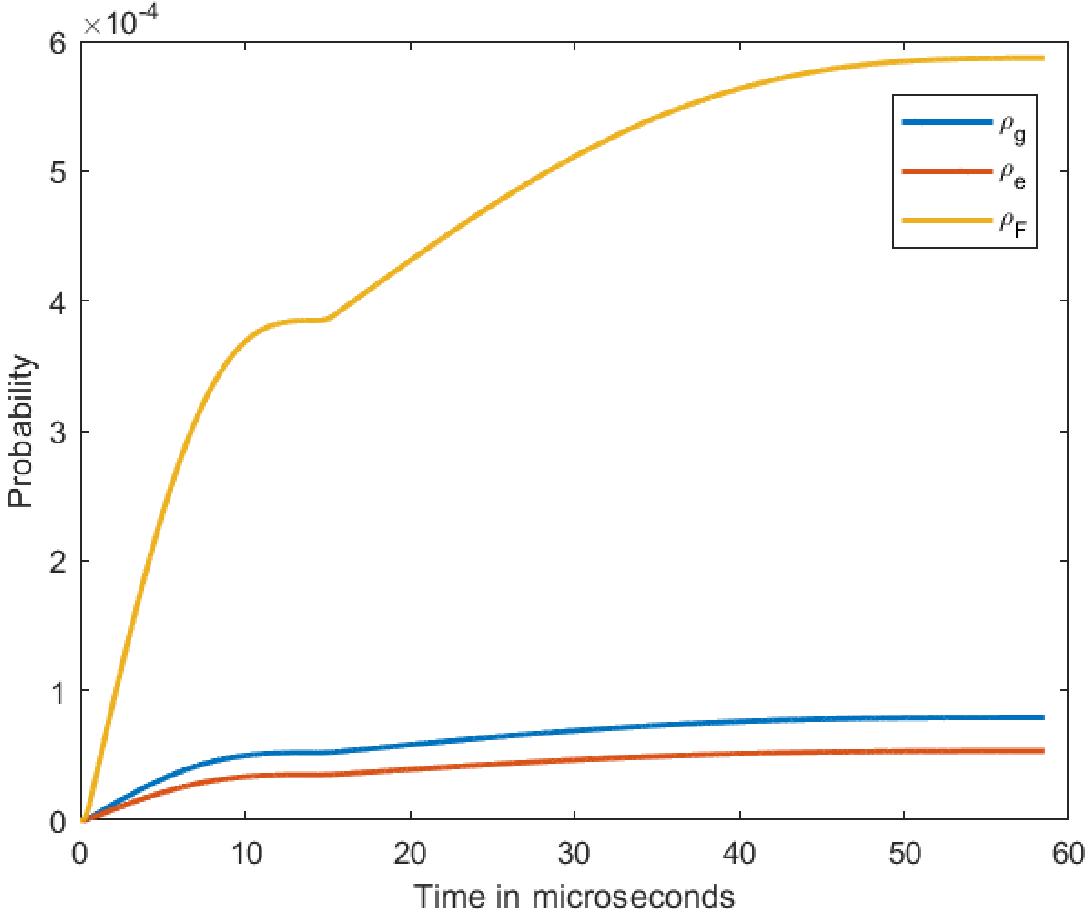
[5]Tan *et al*. Phys. Rev. A, **104**, L010802 (2021)

Simulating an unsharp measurement with decay

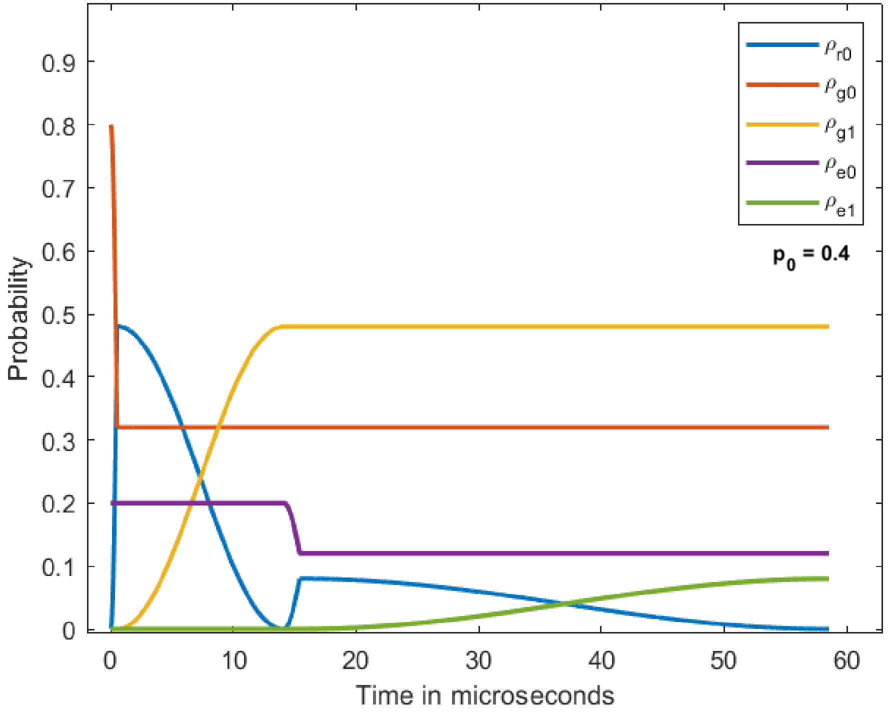
The population evolution of the 5 energy levels during the 4-step procedure



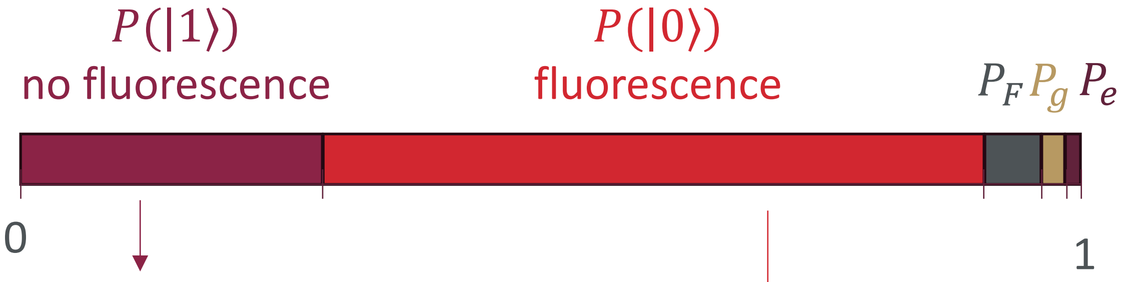
The accumulated spontaneous decay



Measurement on auxiliary ion



- The following step is a projective measurement on the auxiliary ion done by electron shelving and fluorescence detection.
- The measurement result is giving by the probabilities for the different processes:



$$\psi_{\text{target, none}} = c_1\sqrt{p_0}\,|g_t\rangle + c_2\sqrt{1-p_0}\,|e_t\rangle$$

$$\psi_{\text{target, fluoro}} = c_1\sqrt{1-p_0}\,|g_t\rangle + c_2\sqrt{p_0}\,|e_t\rangle$$

$$\psi_{\text{target, decay}} = |F_{7/2}\rangle$$

$|g_t\rangle$

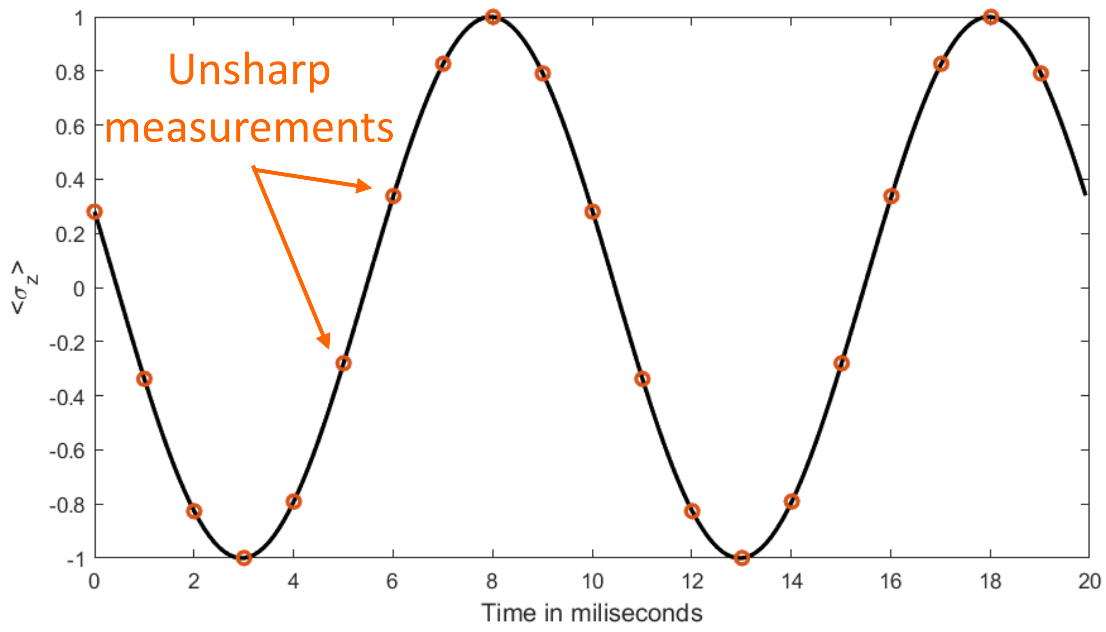
$|e_t\rangle$

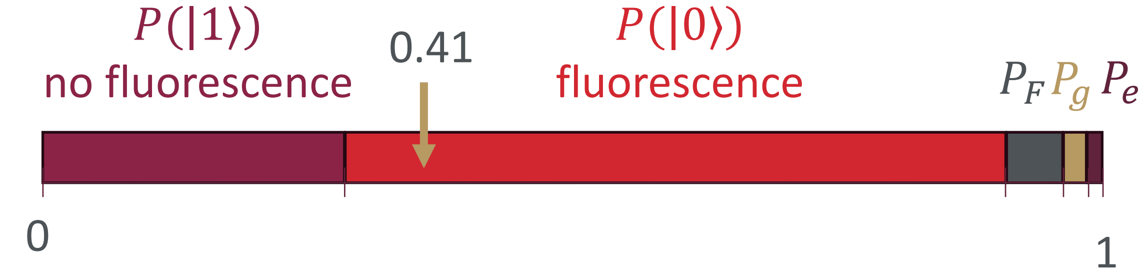
Simulating the qubit state

1. Time evolution - Rabi

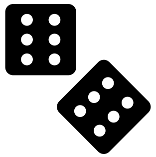
2. OB simulation determines: probabilities for the 5 outcomes of fluorescence, none, and three spontaneous decays.

3. Roll the dice and update the state accordingly then repeat





$$\psi_{\text{target, fluoro}} = c_1\sqrt{1-p_0}\,|g_t\rangle + c_2\sqrt{p_0}\,|e_t\rangle$$



The estimation scheme

Qubit state evolution

1. Time evolution - Rabi

2. OB simulation determines: probabilities for the 5 outcomes of fluorescence, none, and three spontaneous decays

3. Roll the dice and update the state accordingly then repeat

Estimate state evolution

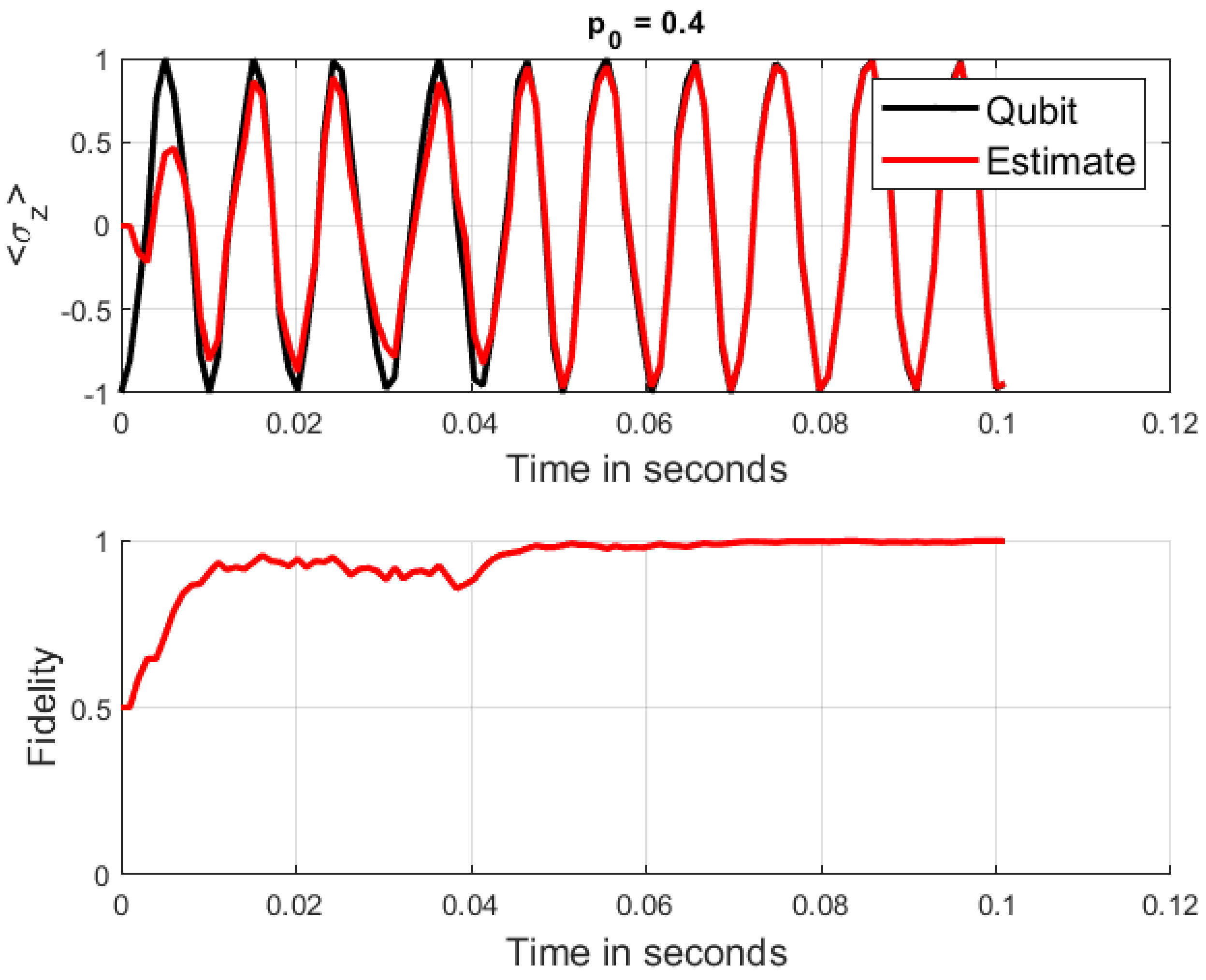
Initialise with a random state

1. Time evolution - Rabi

2. Use the measurement outcome but only for fluorescence or none. Update the state accordingly then repeat

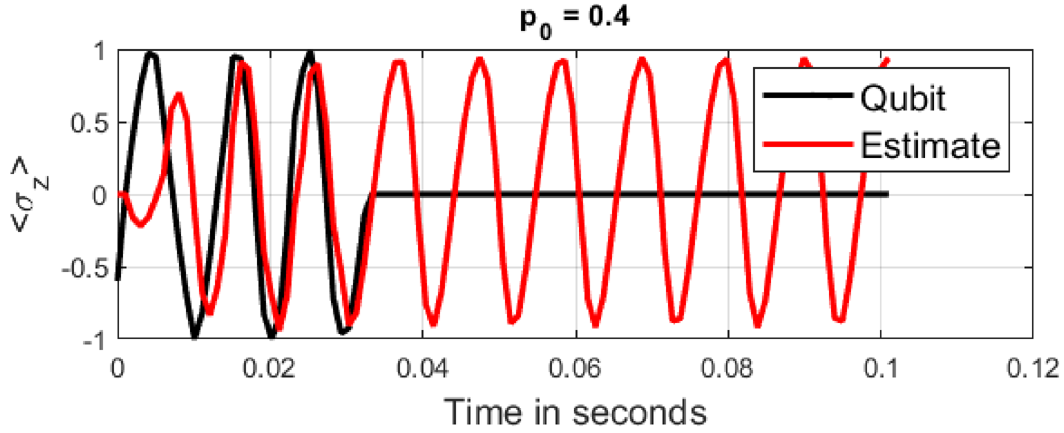
The estimation scheme

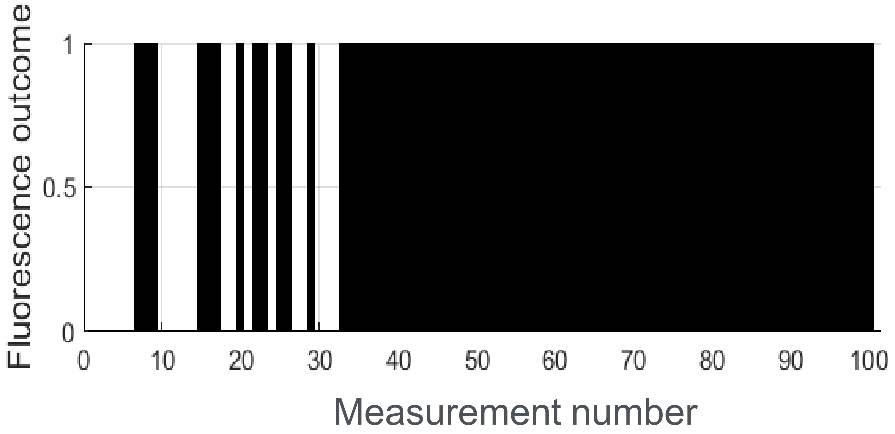
- 10 unsharp measurements are made per Rabi cycle
- $\Omega_{\text{qubit}} /(2\pi) = 100$ Hz



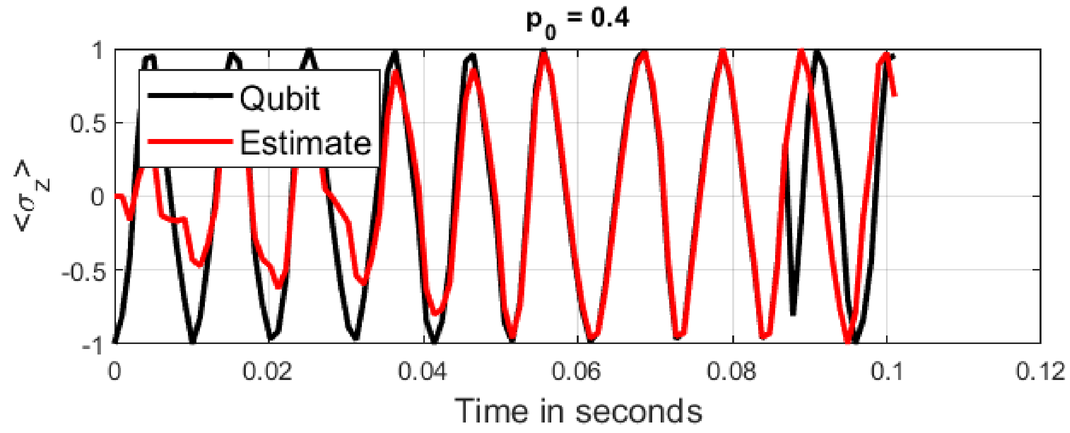
Spontaneous decay in the estimation scheme

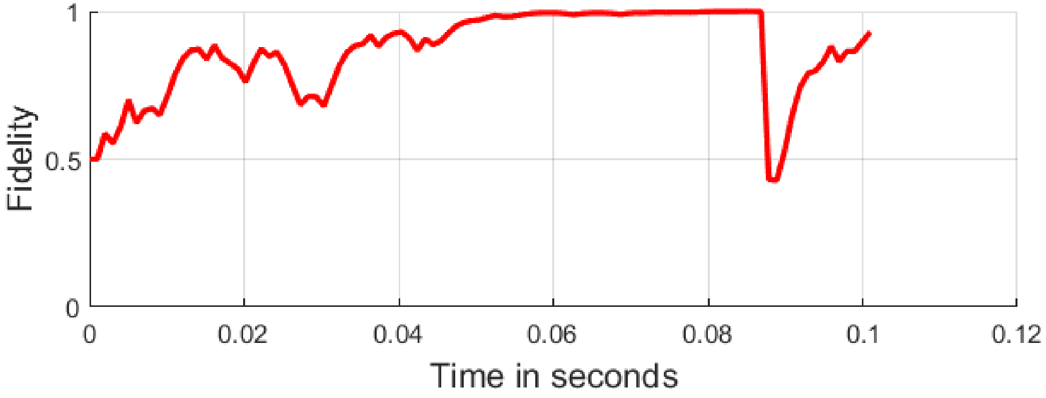
Decay into the ${}^2F_{7/2}$ level (dark state)





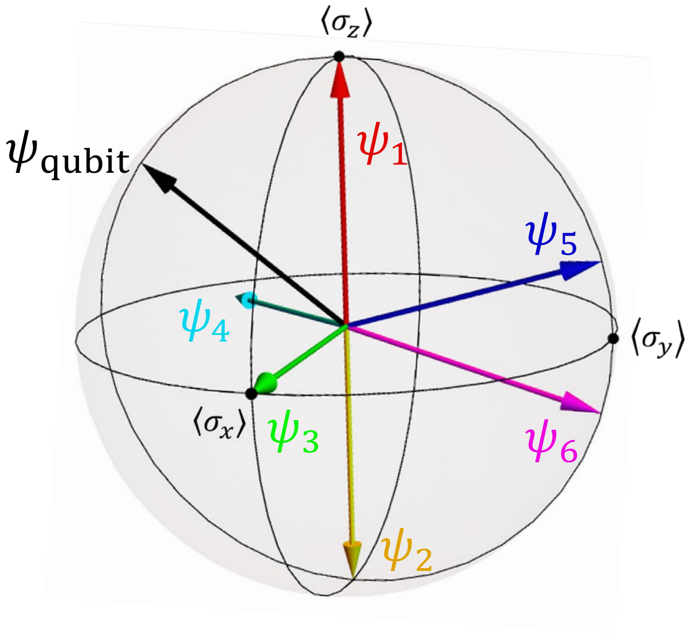
Decay into the ${}^2S_{1/2}$ level (into one of the qubit states)

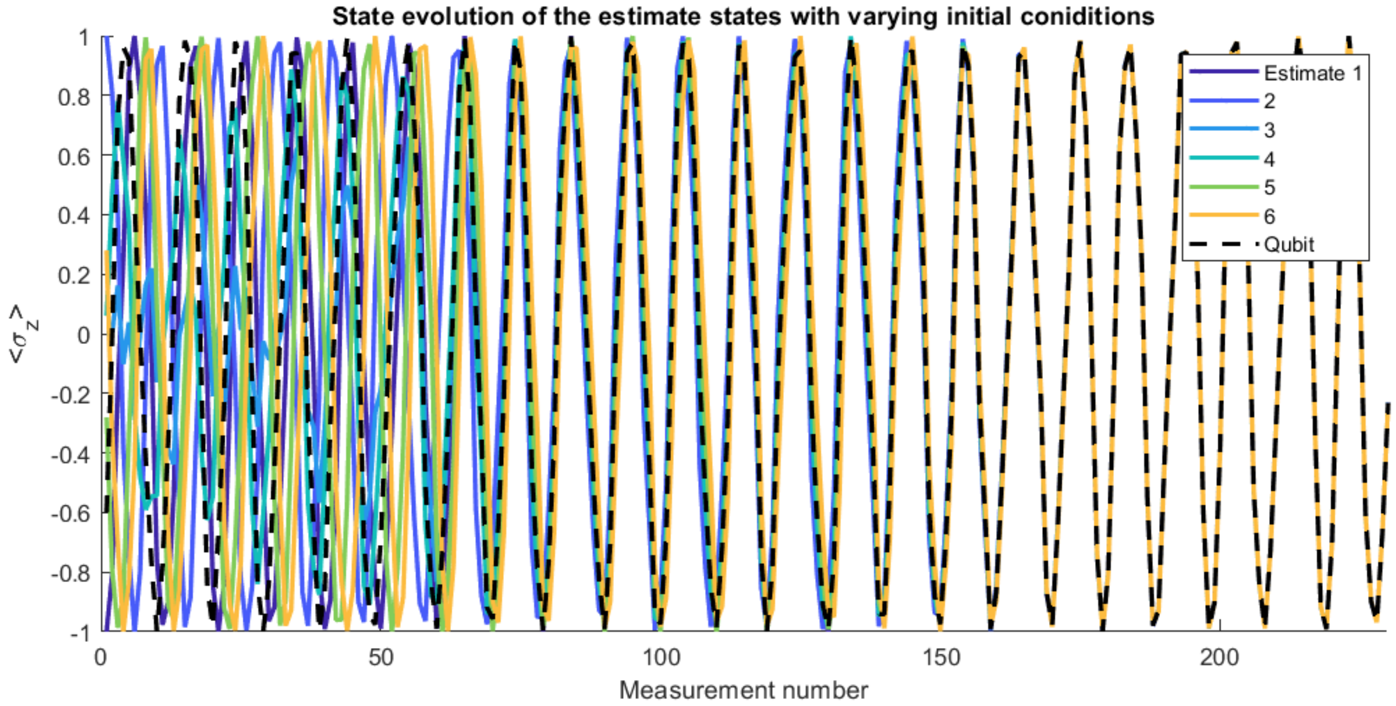




Convergence protocol: Multi-initialisation technique

The initial states on the Bloch sphere





The initial estimate and qubit states

$\psi_1 = 1|g_t\rangle + 0|e_t\rangle$

$\psi_2 = 0|g_t\rangle + 1|e_t\rangle$

$\psi_3 = \sqrt{0.5}\,|g_t\rangle + \sqrt{0.5}|e_t\rangle$

$\psi_4 = (0.342 + 0.5942\, i\,)|g_t\rangle + (0.2 - 0.7i\,)|e_t\rangle$

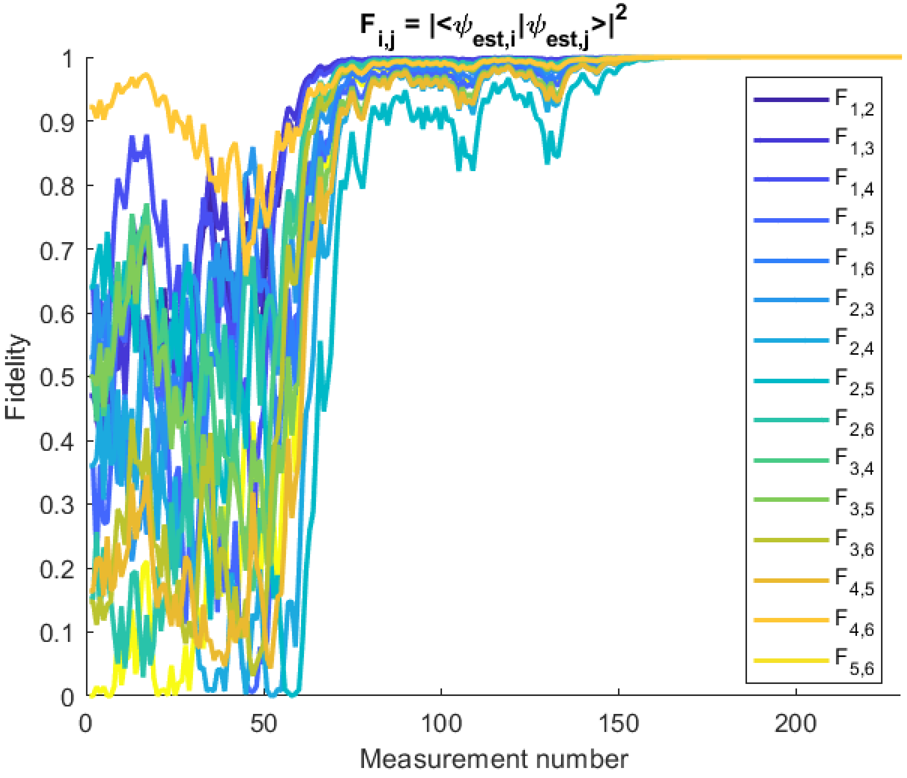
$\psi_5 = 0.8\,|g_t\rangle + 0.6\, i\,|e_t\rangle$

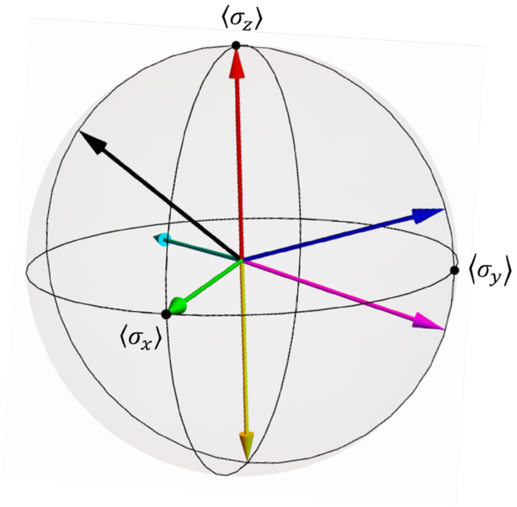
$\psi_6 = 0.6\,|g_t\rangle + 0.8\, i\,|e_t\rangle$

$\psi_{\text{qubit}} = \sqrt{0.8}\,|g_t\rangle - \sqrt{0.2}\, i\,|e_t\rangle$

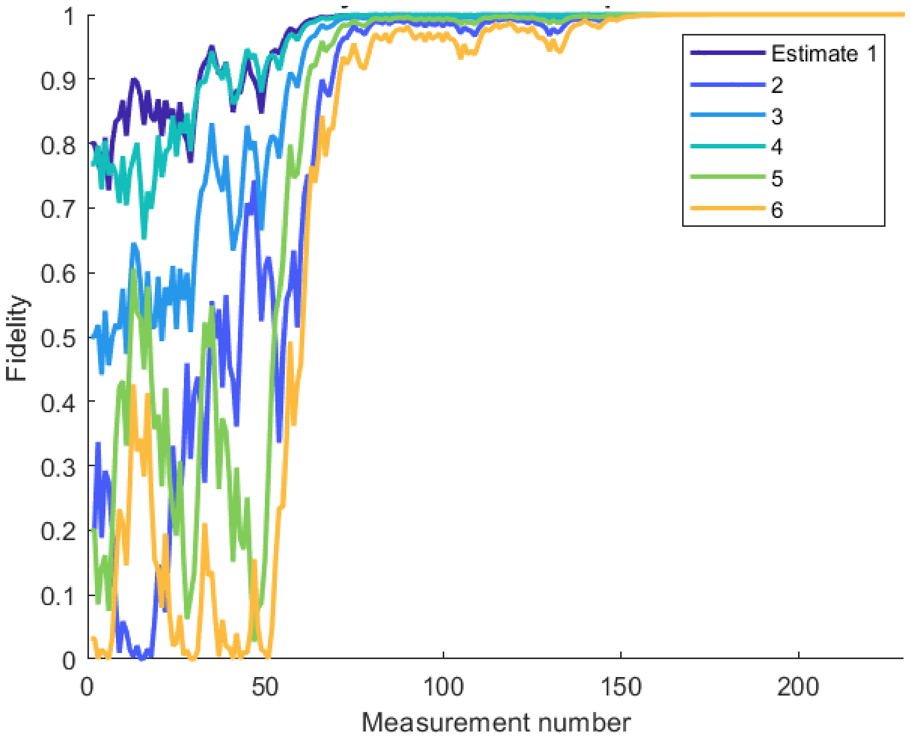
Convergence protocol: Pairwise estimate fidelity

Fidelities between each pairwise combination of estimate states





Fidelities between each estimate and the qubit state



Simulating unsharp measurements and state estimation using trapped ytterbium ions

Conclusions

- We **simulated tracking Rabi oscillations** of a trapped ytterbium ion qubit probed by unsharp measurements.
- We identified a **practical convergence protocol** for the estimation scheme which does not rely on knowledge of the qubit evolution.
- The Bloch model gave insight on the **timescales of qubit dynamics** which are possible to study using unsharp measurements.
  - Slow qubit dynamics $\sim 10 - 100$ ms when using continuous lasers.
  - Alternative implementations using pulsed lasers and Raman transitions[6] could speed up these processes.

[6] Campbell *et al.* Phys. Rev. Lett **105**, 090502 (2010)

General conclusions

Perspective

- Experimental research in our context only makes sense in combination with theoretical research.
- The experimental perspective and considerations can provide an edge to theoretical research in this field.
- Collaboration is important!

Shaping the future of Physics Report → Isolated projects, limited collaboration → WORKSHOP Quantum Science and Technology across Africa → Collaboration for critical mass and impact?

thank you | enkosi | dankie

Photo by Stefan Els

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